

Dear Editor and Reviewers,

Thank you for the reviewers' valuable comments and suggestions on our manuscript. We have carefully revised the manuscript and uploaded the revised version together with a detailed point-by-point response to all reviewer comments.

Sincerely,
Meliana and her co-authors

Response to Reviewer 1

Manuscript Title: Orchestrating Value Co-Creation in Digital Education Using the TISE-VALORIZE Framework: From Knowing to Becoming

Manuscript ID: 1827542

Article Type: Original Research

Journal: Frontiers in Education - Digital Learning Innovations

Comment 1

The study presents a timely and original framework that rethinks engineering education for an AI-driven world by integrating cognitive science, motivation theory, professional identity, and value creation. Its major strengths are strong conceptual novelty, practical classroom mechanisms such as knowledge maps and simulations, and promising preliminary evidence of higher and more consistent student performance. However, studying also has notable weaknesses. The empirical validation is still limited by a single-institution, cross-cohort design, modest effect size, insufficient direct measurement of motivation and cognitive load, and possible confounding across semesters. Thus, the framework is promising, but stronger large-scale validation is still needed.

Response:

We sincerely thank the reviewer for this thoughtful and balanced assessment of our manuscript. We appreciate the reviewer's recognition of the framework's conceptual originality, its practical classroom mechanisms, and the preliminary evidence of higher and more consistent student performance observed in the intervention group.

We also agree with the reviewer's concerns regarding the current empirical scope of the study. In response, we revised the manuscript to make these limitations more explicit and to align our interpretation more carefully with the available evidence. The revised manuscript now states more clearly that the present study evaluates only the performance-related dimension of the framework, whereas the proposed mechanisms related to motivation, cognitive load, professional identity, and transfer were not directly measured in the current dataset and therefore remain propositions for future empirical testing.

We also strengthened the methodological description of the study. The manuscript now explicitly describes the empirical component as a posttest-only quasi-experimental design with non-equivalent cross-cohort groups in a single institutional context, and it states more clearly that the findings should be interpreted as preliminary comparative evidence rather than causal proof. In addition, the Results and Discussion sections now acknowledge more directly that the observed effect size was modest, and that cohort- or semester-level confounding cannot be ruled out.

Finally, we revised the Limitations and Future Directions sections to emphasize the need for stronger future validation, including more rigorous designs, larger and more diverse samples, and direct measures of the proposed cognitive, motivational, and identity-related mechanisms. We believe these revisions improve the manuscript's transparency and make its claims more consistent with the evidence currently presented.

Changes made in the manuscript:

The manuscript was revised to clarify the preliminary empirical scope of the study, explicitly acknowledge the absence of direct measures for motivation, cognitive load, professional identity, and transfer, describe the design more cautiously, interpret the modest effect size more carefully, and strengthen the limitations and future research agenda.

- Preliminary empirical scope clarified: Introduction, pages 2–3, lines 64–92.
- Design limitations stated more explicitly: Section 5.1–5.4, pages 10–11, lines 333–369.

- Modest effect size interpreted cautiously: Section 5.5 Results, page 12, lines 383–391.
- Semester/cohort confounding acknowledged: Section 6.3 Limitations, page 15, lines 455–465.
- Need for stronger future validation emphasized: Section 6.4 Future Directions and Section 7 Conclusion, page 15, lines 466–481.

Comment 2

Sample Size not mentioned (Abstract)

Response:

We thank the reviewer for this helpful comment. In response, we revised the Abstract to explicitly report the total sample size and the group composition. The revised abstract now states that the study involved 138 undergraduate engineering students, consisting of one intervention group implementing TISE-VALORIZE ($n = 53$) and two control groups receiving conventional instruction ($n = 42$; $n = 43$). This revision was made to improve methodological clarity and provide a clearer overview of the empirical scope of the study.

Changes made in the manuscript:

The Abstract was revised to explicitly include the total sample size and the number of participants in each group.

- Sample size added explicitly in the Abstract: Abstract, page 1, lines 15–18.

Comment 3

Discussions and Implications are not mentioned

Response:

We thank the reviewer for this helpful comment. In response, we revised the manuscript to make the discussion and implications more explicit. Specifically, we strengthened the Discussion section to clarify the theoretical implications of the TISE-VALORIZE framework, its practical implications for course redesign in digital engineering education, and the boundaries of interpretation of the present findings. We also added clearer statements on the study's limitations and future research directions to ensure that the implications are presented in a balanced and evidence-based manner. In addition, the Abstract and Conclusion were revised to include more explicit implication-oriented statements while maintaining a cautious interpretation consistent with the scope of the current evidence.

Changes made in the manuscript:

Revisions were made in the Abstract, Section 6 (Discussion: Theoretical Implications and Addressing Counterarguments), and Section 7 (Conclusion) to explicitly state the discussion points, practical and theoretical implications, limitations, and future research directions.

- Implication-oriented statement added in the Abstract: Abstract, page 1, lines 25–29.
- Discussion section revised and organized explicitly: Section 6, pages 14–15, lines 426–471.
- Practical and theoretical implications clarified: Section 6, pages 14–15, lines 427–454.
- Limitations and future directions stated explicitly: Section 6.3–6.4, page 15, lines 455–471.
- Conclusion revised to include practical implications: Section 7, page 15, lines 473–481.

Highlights Version

Manuscript Title: Orchestrating Value Co-Creation in Digital Education Using the TISE-VALORIZE Framework: From Knowing to Becoming

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Orchestrating Value Co-Creation in Digital Education Using the TISE-VALORIZE Framework: From Knowing to Becoming

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10 **Keywords:** artificial intelligence, cognitive load theory, engineering education, professional
11 identity formation, self-determination theory, TISE-VALORIZE framework, value co-creation

12 **Abstract**

13 AI can automate technical activities, but it can't teach skills that only people have. This study proposes
14 the TISE-VALORIZE framework and presents a preliminary empirical examination of its association
15 with student performance outcomes in digital engineering education. A posttest-only quasi-
16 experimental study with non-equivalent cross-cohort groups involved 138 undergraduate engineering
17 students: one intervention group implementing TISE-VALORIZE ($n = 53$) and two control groups
18 receiving conventional instruction ($n = 42$; $n = 43$). Student performance was evaluated through
19 Structured Academic Activities assessed with Bloom's Taxonomy-aligned rubric. The intervention
20 group showed higher mean scores ($M = 94.98$, $SD = 7.21$) and lower score variance ($Var = 52.01$) than
21 Control 1 ($M = 88.70$, $SD = 18.33$, $Var = 335.78$) and Control 2 ($M = 86.54$, $SD = 17.05$, $Var = 290.98$).
22 A one-way ANOVA indicated a significant group difference, $F(2, 135) = 4.41$, $p = .014$, $\eta^2 = .061$, and
23 Tukey post-hoc analysis showed a significant difference between the intervention group and Control
24 2, while the difference between the intervention group and Control 1 did not reach significance. These
25 preliminary results suggest that the framework may support more consistent learning outcomes.
26 However, direct measures of cognitive load, motivation, and related psychological processes were not
27 included in the present study. These findings provide preliminary support for the framework's potential
28 to improve performance consistency, while larger-scale studies with direct measures of cognitive and
29 motivational processes are needed.

30 **1 Introduction**

31 Engineering education is shifting from a content-based approach to competency development
32 (Charytanowicz, 2023). Other empirical studies indicate that the disparity between education and
33 industry endures, even among high-achieving graduates: analyses of software engineering practices
34 consistently reveal a misalignment between the focus of higher education programs and the
35 requirements of practitioners in professional settings, including both technical and professional aspects
36 (Garousi, Giray, and Tuzun, 2019). To build a long-lasting smart campus ecosystem, many tools and

frameworks are being created to test preparedness and include digital transformation (Imbar *et al.*, 2025). Engineering education has long emphasized foundational knowledge, routine problem solving, and performance assessment. Students study the basics, figure out how to tackle frequent situations, and test their skills with tests. High grades showed that you were ready to work in the field. But this approach included an unspoken assumption: that the mental effort needed to solve engineering issues was the same as the mental labor needed to be an engineer. This assumption is no longer valid.

Learning requires learners to organize information meaningfully and integrate it into schemas stored in long-term memory. (Sweller, Van Merriënboer, and Paas, 1998). One important thing about working memory, which is a mental region where new knowledge is kept, is that it can only contain roughly 3–7 bits of information at a time. Unless actively rehearsed, such information typically decays within 15–30 seconds (Miller, 1956). Students will have trouble learning and developing their abilities if the environment has too much information in it. Cognitive Load Theory (CLT) offers a solution: instructional design should minimize extraneous cognitive load to optimize working memory for effective learning termed 'germane load' by CLT researchers. CLT serves as a cognitive framework for crafting educational experiences that cultivate robust skills (Robins, 2022). It must alleviate superfluous cognitive burden to liberate working memory for authentic learning, and it must structure teaching to facilitate the development of meaningful, practical knowledge frameworks that embody professional competence. It must also do this in a manner that respects autonomy, competence, and relatedness that keep people interested. CLT posits that the complexity of a topic correlates with the interaction of its components. Darejeh, Marcus, and Sweller (2021) examined the impact of narrative style on cognitive load during the acquisition of a difficult productivity software program. The researchers used two categories of materials, distinguished by low and high element interaction, according to each story style.

The theoretical framework delineated below aims to actualize this possibility. It is based on well-known learning theories (Cognitive Load Theory (CLT), Self-Determination Theory (SDT), and research on expertise) and considers the realities of modern technology (AI integration, digital tools, and participatory design). Its goal is to create professionals who can lead and make a difference in a world that is changing quickly. In this preliminary study, we empirically evaluate only the performance-related dimension of the framework, while the motivational, cognitive-load, transfer, and professional-identity mechanisms remain theoretical propositions requiring direct measurement in future work. The framework draws on Cognitive Load Theory, Self-Determination Theory, and professional identity formation because these perspectives address three complementary dimensions of engineering education. Cognitive Load Theory explains how structured learning design and external representations may support schema construction and reduce unnecessary processing demands. Self-Determination Theory explains how autonomy, competence, and relatedness may support sustained engagement in learning activities. Professional identity formation helps explain why professional simulation and value-oriented participation matter in engineering education beyond content mastery alone. Together, these perspectives provide the theoretical basis for the TISE-VALORIZE framework and for its expected influence on both learning processes and performance outcomes.

In the present preliminary empirical study, two hypotheses were examined directly:

- H1. Students in the TISE-VALORIZE group will achieve higher Structured Academic Activities scores than students in the conventional instruction groups.
- H2. Students in the TISE-VALORIZE group will show lower score dispersion than students in the conventional instruction groups.

In addition, the framework proposes several secondary hypotheses for future empirical testing:

- H3. The framework may reduce perceived cognitive load through structured knowledge mapping and scaffolding.
- H4. The framework may support intrinsic motivation and learning agency through autonomy-supportive and contribution-oriented learning activities.
- H5. The framework may support stronger professional identity formation and learning transfer through professional simulation and value-oriented participation.
- H6. The framework may increase the quality and reuse value of student-produced artefacts within the learning community.

Because the present study did not include direct measures of cognitive load, motivation, professional identity, or transfer, only H1 and H2 are evaluated empirically in this paper.

2 Theoretical Foundations: Merging Cognitive Science and Pedagogy

2.1 Fundamental Ideas About Education

The framework is based on several interconnected ideas that are meant to meet the needs of the AI future. First, Value-Oriented Education (VOE). In traditional education, teachers teach technical information, and students learn it. VOE alters this goal. Higher education should teach how to be good people, how to be professionals, and how to give back to society. Students learn how to develop code and why technological choices might hurt people or the environment, for example. Second, Value Co-Creation (VCC). This word originates from marketing and service writing and talks about how businesses and customers work together to produce value. In education, VCC means that students do not merely receive knowledge passively; they actively contribute to creating learning experiences that matter. They collaborate with educators and technology to develop resources such as knowledge maps, problem-solving tools, and educational aids for their classmates and future learners. In this way, students become co-creators instead than merely consumers of knowledge (Pralhad and Ramaswamy, 2004; Grönroos and Voima, 2014; Robayo-Pinzon *et al.*, 2023).

Knowledge Management (KM) is a broad way for an organization to gather, develop, protect, share, and use knowledge gained from its experiences (Mohammed, 2021; Deev and Finogeev, 2023). It works well with the focus of VOE and VCC. Collaborative Learning (CL) is a strategy in which a small group of students work together to achieve a shared objective or complete an assignment (Ouyang, Xu and Cukurova, 2023; Ramadevi *et al.*, 2023). Lastly, Personalized Learning (PL) is a method of teaching that changes based on how fast students learn and how lessons are given to help them achieve their best (Labib, Canós, and Penadés, 2017; Deev and Finogeev, 2023; Apirating, 2025).

2.2 Psychological and Cognitive Foundations

CKMS-SE combines old ways of thinking about psychology and cognition:

- Self-Determination Theory (SDT) explains that individuals are intrinsically driven when their demands for autonomy, competence, and relatedness are satisfied (Weydner-volkmann and Bär, 2024; Annamalai and Nasor, 2025).
- Cognitive Load Theory (CLT) emphasizes that successful learning is facilitated by reducing irrelevant cognitive burden (extrinsic load) and enhancing cognitive load that aids the learning process (Alsaffar, Alfayly, and Ali, 2022; Apirating, 2025).

- People that have similar aims, interests, and skill levels make up Communities of Practice (CoP). They acquire information collaboratively by active participation and contribution as a cohesive unit (Tsang and Tsui, 2017; Taft, 2019).

2.2.1 Self-Determination Theory and Intrinsic Motivation

CLT elucidates the cognitive frameworks integral to the learning process, while SDT emphasizes the motivating factors that sustain student engagement and facilitate profound learning (Ninghardjanti *et al.*, 2025). SDT says that human motivation exists on a scale. At one end is external regulation, where people do things because they feel pressure or get benefits from outside sources. At the other end is integrated regulation, where people do things that are in line with their own beliefs and identities. SDT also makes a distinction between the amount of motivation (how strongly someone feels motivated) and the kind of motivation (the qualities or categories of motivation they have).

Students who still use conventional learning approaches tend to depend on grades or test results as rewards. They usually use shallow learning tactics, stop when the benefits stop, and say they are less happy. Conversely, learners driven by intrinsic factors—such as curiosity, autonomy, and the aspiration to make a significant impact, employ more profound learning strategies, maintain persistence despite the lack of external incentives, and experience enhanced well-being and continuous engagement.

2.2.2 Cognitive Load Theory and Schema Construction

New information is first processed in working memory. Working memory has severe capacity and duration constraints. (Duran, Zavgorodniaia, and Sorva, 2022). Learning is the process of transferring information from working memory to long-term memory, where it is organized into mental schemas that integrate multiple elements into a coherent whole. The creation and automation of such schemas are central to expertise development (Robins, 2022).

This research established the VALORIZE strategy, which aims to design learning that alleviates the three forms of cognitive burden on students and enhances information processing in their working memory, drawing on cognitive load theory (CLT) and dual coding theory (DCT). Students convert the signal and system domains, which were initially expressed in words and arithmetic equations, into knowledge maps and problem-solving maps.

The following design principle is clear: to make learning as effective as possible, instructional designers should reduce extraneous load, manage intrinsic load correctly (by providing scaffolding for difficult material), and increase germane load, which is the amount of working memory used for productive learning activities (Duran, Zavgorodniaia, and Sorva, 2022).

2.2.3 Community of Practice: Professional Identity and Expertise Development

The idea of professional identity, which is taking on the responsibilities, attitudes, and behaviors that come with a job, is an important part of the framework. Professional identity isn't just something you get by learning new things. It develops over time as you do real professional work, get advice from experienced professionals, and gradually adopt professional standards and values (Pischetola, 2021; Shal *et al.*, 2026).

Moreover, ethical thinking and responsible professional judgment are fundamental to professional identity (Große-Bölting *et al.*, 2023). Engineers do not work in a vacuum; their choices have an impact on people's health, the environment, and social justice (Cortázar *et al.*, 2024). A professional identity

must include a dedication to ethical ideals (Szynkiewicz, 2025) and the ability to maneuver through intricate scenarios where technical and ethical factors converge (Johannsen, 2026).

3 The TISE Architecture: Operationalizing Multifaceted Intelligence

In this paper, TISE-VALORIZE refers to the overarching theoretical framework, while CKMS-SE refers to its system-level implementation architecture. The suggested way to address the educational gap is the Collaborative Knowledge Management System for Smart Engineering (CKMS-SE). CKMS-SE turns the TISE-VALORIZE theory into a real way to solve problems that use human, artificial, and natural intelligence in students' everyday learning activities. The approach is based on dual-engine architecture. The first engine, called the PUDAL engine, organizes students' thinking into repeated cycles of Perceive–Understand–Design–Act–Learn (PUDAL). The second engine, called the PSKVE engine, manages a multi-asset value portfolio (Product, Service, Knowledge, Values, Environment) to find and reward multi-dimensional value co-creation.

3.1 Defining the Triune-Intelligence Model

Triune-Intelligence Smart Engineering (TISE) proposes a method for intelligent problem-solving that integrates three complementary forms of intelligence: Human intelligence distinguishes individuals by their capacity for moral reasoning, alignment of values, strategic thinking, innovative solution generation, and comprehension of contextual nuances. Artificial intelligence is the capacity to swiftly see patterns, make things better, put together a lot of knowledge, and follow computer instructions that are clear. AI systems are fantastic at working with large amounts of data, seeing minor patterns, and repeating the same tasks again with remarkable precision. Natural Intelligence: The knowledge that comes from the rules of society, the way things work in nature, and the way people think they should be. This kind of intelligence involves learning about the rules and patterns of nature, how various civilizations make sense of things, and the information that is passed down via different groups and traditions. It makes sure that new technologies are good for people and the environment in the long run.

3.2 The PUDAL Thinking Cycle as A Cognitive Method

The PUDAL engine has five steps that make up a full cycle: Perception (P): Students find stakeholders, data sources, and environmental limitations that are important to the issue. Understanding (U): They use the values at play, hidden patterns, and physical laws to get a full picture of what's going on. Solution Development (D): They look at more than one option, not simply the best one. Action (A): They put the selected solution into action by making a physical item, such code, an analytical report, a user interface, or a prototype. Learning (L): They think about how what they do affects people, computers, and the environment.

VALORIZE employs Knowledge Maps structured in a multi-graph taxonomy to make this expert cognitive process clear. This is like a "grammar of knowledge" that helps experts think. Formal knowledge artifacts back up each step of PUDAL (see Table 1). The whole process isn't a straight line; instead, the PUDAL cycle is iterative and always changing, like micro-evolution that leads to constant progress.

Table 1. PUDAL Cycle: Analysis of TISE and Artefact Requirements

PUDAL Phase	Human Intelligence Analysis Task	Artificial Intelligence Analysis Task	Nature Intelligence Analysis Task	Required VALORIZE Artefacts
Perceive (P)	Find out who the most important stakeholders are and what they think about the problem in general.	Define data sources and collection strategies.	Find the most important environmental indicators and physical limits that need to be assessed.	Factual Association Map
Understand (U)	Find out what stakeholders' value, what ethical challenges they face, and what the cultural environment is.	Analyze data to identify patterns, correlations, and barriers.	Model the laws of physics and the rules of nature that control the system.	Portfolio: Directed Acyclic Graph (DAG), Decomposition Tree, Flowchart
Decision-making and Planning (D)	Set up ethical guardrails and success metrics that are based on people.	Use optimization algorithms or simulations to evaluate solution options.	Look at your possibilities in light of your resources and how well they work.	Trade-Off Analysis Map (Weighted Graph)
Act-Response (A)	Make user interfaces and service delivery models that people can utilize.	Make algorithms for controlling things, pipelines for processing data, or actions for robots.	Design physical components and artefact mechanisms.	Process Application Map (Flowchart)
Learning-Evaluating (L)	Think about how the answer will affect people's health and equality in the long run.	Think about how the answer will affect people's health and equality in the long run.	Evaluate the non-negotiable final impact of the solution on the environment.	Synthesized Design Map (Network Graph)

203 **3.3 The PSKVE engine as a value-creation method**

204 The Knowledge Marketplace was changed into a multi-asset PSKVE value portfolio (see Table 2).

205 **Table 2. Dimensions of PSKVE and TISE-VALORIZE Assets**

PSKVE Dimension	TISE-VALORIZE Asset	Dimension of Valued Worth	Example of Artefact Value Creation
Product (P)	Productivity Units (PU)	Economic Value (Productivity and Efficiency)	Writing Python programs to make computations easier or improve designs.
Service (S)	Service Vouchers (SV)	Human-Centered Design (Well-being and Empathy)	Do a thorough study of stakeholders or a map of the user experience.
Knowledge (K)	Knowledge Credits (KC)	Cognitive Depth (Intellectual Capacity)	Makes a formal Directed Acyclic Graph (DAG) that shows how different concepts are related to each other.
Value (V)	Economic Shares (ES)	Sustainable Exchange (Financial/Social Viability)	Come up with a workable business model or cost-benefit analysis for fair value distribution.
Environment (E)	Sustainability Bonds (SB)	Protective/Ethical Values (Risk and Sustainability)	Do a life cycle evaluation or create systems that really reduce their influence on the environment.

206 **3.4 Operationalization through the Knowledge Marketplace and Knowledge Maps**

207 The method uses two main tools to embed the dual engines into daily learning: (1) a knowledge
208 Marketplace, where lecturers post weekly "learning needs" as sprint goals and students voluntarily
209 respond by making PSKVE-typed artefacts; and (2) AI-assisted Knowledge Maps will help students
210 express their thoughts using the formal multi-graph taxonomy, like Directed Acyclic Graphs (DAGs),
211 flowcharts, and weighted graphs. Git and GitHub make it simple to maintain track of all their assets
212 and make sure that contributions are clear, can be used again, and are straightforward to verify.

213 4 VALORIZE Learning: A Participatory Pedagogy for Professional Formation

214 4.1 Core Principles and the Change from Knowing to Being a Professional

215 VALORIZE fundamentally shifts the focus of education from content mastery, which is learning and
216 repeating facts and procedures, to professional formation, which is developing the intellectual, ethical,
217 and social traits that make a person a good, responsible professional. This change isn't only a change
218 in words. It has big effects on how teachers and students are responsible for their own learning, how
219 they teach, and how they grade. In a content-based learning model, pupils are seen as individuals who
220 acquire knowledge. The curriculum includes things that need to be taught, and the teachers are
221 professionals who offer that knowledge. How effectively pupils can repeat that material is how success
222 is measured. In a professional development model, on the other hand, students are seen as future
223 professionals who are doing real work. The curriculum includes things that happen in the workplace,
224 and teachers act as mentors to help students grow professionally. Success is based on the ability to add
225 value and contribute.

226 4.2 Professional Simulation as Pedagogical Strategy

227 The lecturer's job varies a lot. The lecturer is no longer merely a "sage on the stage" who shares
228 information. Now, they are "representative and role models of the profession." This isn't just a shift in
229 words. There are real-world effects: the lecturer demonstrates how to handle difficulties like a pro by
230 stating their ideas out loud and making their unspoken knowledge evident. The teacher teaches students
231 how to be professionals by providing them feedback that goes beyond merely being accurate and looks
232 at aspects like how well they match with professional standards, quality, and depth.
233 Professional values are not only an additional feature of the VALORIZE paradigm; they are the most
234 significant part of any learning activity. When students create control systems, they must not only
235 improve their technical performance but also consider their impact on energy consumption, user-
236 friendliness, and the environment. Learning sends a clear message: being good at something and being
237 ethically responsible are two sides of the same coin.

238 4.3 The Knowledge Marketplace: An Economic Model for Motivation and Value Creation

239 The knowledge map is a mental structure that helps people learn, while the Knowledge Marketplace is
240 a social and motivating framework. This marketplace changes the way students, teachers, and
241 knowledge interact with each other. Instead of just following rules and taking in information, they are
242 now creating and sharing it.

243 4.3.1 How it works and why

244 The Knowledge Marketplace functions like a real marketplace. Teachers publish "learning needs" as
245 knowledge assignments (such as building a signal-system concept map), and students respond by
246 generating high-quality artifacts, like knowledge maps or problem-solving tools. Then, students may
247 use digital money to "buy" artifacts that match the requirements. These artifacts are enduring resources
248 that future organizations may use to learn from and build on. This sets off a virtuous cycle: when
249 students' effort adds value, they receive higher marks, and their classmates have access to richer
250 learning resources.

251 Teachers need to ask for artifacts first. They normally put them into groups based on how hard they
252 are to think about and the technical field. Then, students make Primitive Maps and Problem-Solving
253 Models that are good for teaching. It takes a lot of time and thinking to become an expert at these.
254 Next, the artifacts are evaluated to see whether they are comprehensive, clear, and in line with what

experts know. People who pass the examinations may "buy" them via a tiered digital currency system. The course's Knowledge Management System releases a few artifacts. These artifacts become permanent resources that present and future students may use and edit, which helps the community learn more. Lastly, pupils' final marks depend on how much money they gain by selling artifacts. This makes it clear that performing well in higher education is directly related to being able to undertake important intellectual work. It also promotes intrinsic motivation (since the activity is regarded as benefiting the community), competency via explicit feedback, and autonomy because students may pick what objects they build and how deep their labor is.

4.3.2 Gamifying Cognition: Aligning Incentives with Bloom's Taxonomy

The Knowledge Marketplace has a tiered currency system that is meant to encourage deeper study. Each level is closely related to Bloom's Taxonomy. At the basic level (Money Points, Levels 1–2), students learn the basics by making clear and correct idea maps and systems. When students reach the application level (Gold Points, Level 3), they must show how these ideas operate in actual engineering projects, such as making a filter for an audio system. They look at numerous ways to solve a problem and talk about the pros and cons of each at the analysis and assessment level (Platinum Points, Levels 4–5). At the highest level (Diamond Points, Level 6), they come up with whole new ways to solve hard and messy technical problems. This tiered framework helps students who want to get the most out of their learning naturally go from memorizing to analyzing and synthesizing.

4.4 Roles And Agency: From Consumers to Co-Designers

4.4.1 The Student as Metacognitive Agent and a Source of Value

Students in VALORIZE have a significantly different role than they do in conventional classes. They don't simply passively take in information; they actively learn and add to the group's knowledge. Creating Knowledge Actively: The major role of students is to make knowledge artifacts, like Primitive and Problem-Solving Maps, that indicate they understand the material and are useful to the course community. This is not simply a section of the test; it's the most important aspect. Assessment occurs not via high-stakes tests, but through the continual evaluation of job quality. Thinking about your own thinking: Students must maintain reflective learning journals that illustrate how they are learning. Using a framework like DAR (Description What Happened; Analysis), students learn to look at their own learning, uncover gaps, and make plans on how to improve. What caused that to happen? What does it mean to reflect? Action Plan What's next? This practice improves metacognitive awareness, which is the ability to think about how your brain works. This is important for being able to adjust. Professional Practice: The framework purposefully incorporates industry-standard protocols into student projects. For example, students may utilize version control systems like Git or GitHub to keep track of their learning artifacts, treating the process of learning as a cycle of development. This serves two purposes: it makes a detailed record of the learning process and teaches students to the professional processes they will use in real life. Portfolio Development: Instead of only being graded on their test scores, students build a portfolio of their greatest work, which they typically share on a personal blog or professional website. This portfolio becomes a real asset that shows prospective employers what you can do and immediately helps you advance in your career.

4.4.2 The Educator as Architect, Mentor, and Community Builder

As Learning Architects, teachers plan the whole learning experience, from planning weekly activities and setting requirements for the Knowledge Marketplace to sequencing tasks to build skills over time and creating social and pedagogical conditions that help students develop their professional identity. This is complicated design job that needs a lot of thought about how people learn. AI takes care of

regular tests (such as giving initial feedback and automated scoring), so teachers have more time and energy to provide good advice. This mentoring is not theoretical; teachers collaborate with students to show them how specialists solve difficult challenges. They ask deep questions to get students to think more deeply, give them feedback that helps them progress, and help each student on their way to become a skilled and moral professional. This design uses two specialized engines that work together to provide a continuous cognitive cycle inside the intelligent engineering paradigm. Figure 1. CKMS-SE System Architecture, which shows the philosophical basis of TISE-VALORIZE, the engines PUDAL and PSKVE that run the system, and the outcomes that were measured. PUDAL engine that puts into action a cycle of constant cognitive progress. This machine is a "micro-evolutionary engine" that pushes constant improvement and change. It is also a PSKVE engine that controls and improves Shared Value Creation (SVC) by making sure that the value provided extends beyond individual learning to the creation of community assets that can be verified. The Core Engine, the AI-based PUDAL Engine for Personalized Learning, and the PSKVE Engine for managing Value Co-Creation (VCC) are the three key parts of this system.

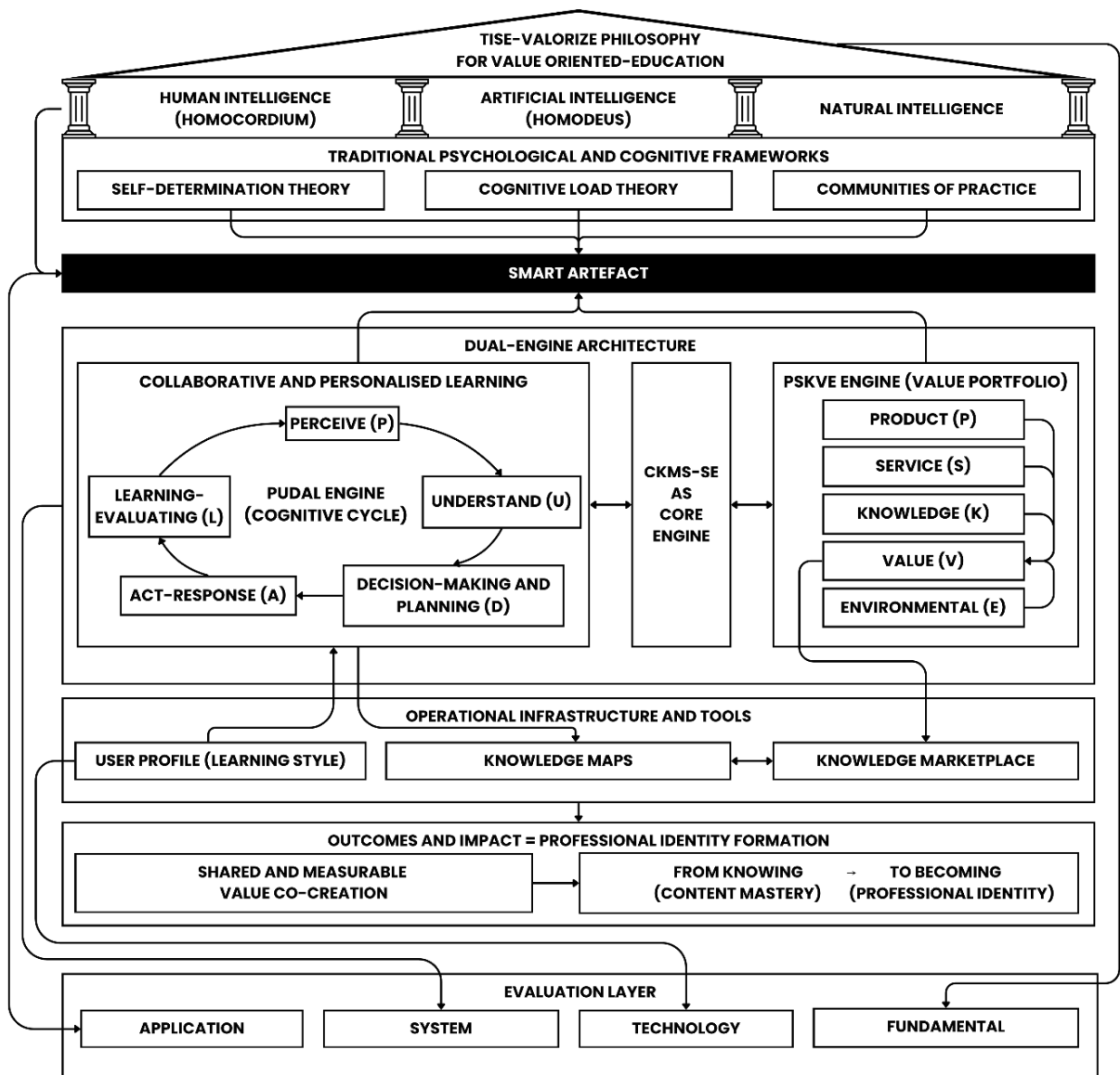


Figure 1. CKMS-SE Architecture

To clarify the empirical scope of the present study, Figure 2 distinguishes between the instructional inputs proposed by the framework, the theoretical mechanisms it draws upon, the performance outcome directly measured in this study, and the broader outcomes proposed for future empirical validation.

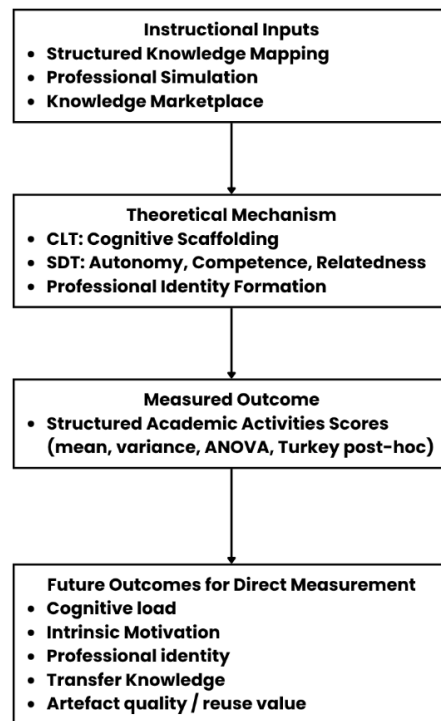


Figure 2. Conceptual scope of the TISE-VALORIZE framework in the present study

5 Preliminary Empirical Validation

5.1 Study Design

The empirical component of this paper employed a posttest-only quasi-experimental design with non-equivalent cross-cohort groups. Random assignment was not feasible because the intervention was implemented in naturally occurring course cohorts across different semesters. Accordingly, the findings should be interpreted as preliminary comparative evidence rather than causal proof.

5.2 Participants and Context

The study involved three cohorts of undergraduate engineering students enrolled in the same course context across different semesters. Control Group 1 (C1) consisted of 42 students from the even semester of 2023/2024 and received conventional instruction. Control Group 2 (C2) consisted of 43 students from the odd semester of 2024/2025 and received conventional instruction. The intervention group (E) consisted of 53 students from the odd semester of 2025/2026 and received instruction using the TISE-VALORIZE framework.

Demographic variables such as age, prior achievement, and access to technology were not systematically collected in a form suitable for retrospective controlled analysis. This limits the extent to which cohort differences can be ruled out statistically and is acknowledged as a limitation of the present study.

5.3 Assessment and Rubric

Student performance was evaluated through Structured Academic Activities using the same Bloom's Taxonomy-aligned rubric across all cohorts. The rubric emphasized conceptual clarity, mapping linkages, procedural reasoning, and integrated synthesis consistent with the expected quality of engineering problem solving. The assessed artefacts included Primitive Knowledge Maps, Problem-Solving Maps, and integrated synthesis artefacts.

Formal psychometric validation and inter-rater reliability statistics for the rubric were not calculated retrospectively in the present study. Therefore, the rubric should be understood as a common instructional assessment instrument rather than a fully validated research-scale measure. This limitation is acknowledged explicitly in the manuscript.

5.4 Statistical Analysis

The analysis included descriptive statistics, one-way ANOVA, Tukey's Honestly Significant Difference (HSD) post-hoc comparisons, Levene's test for homogeneity of variances, and Cohen's *d* for pairwise effect size interpretation. Because no pretest or baseline covariates were collected, ANCOVA could not be applied. Likewise, direct measures of cognitive load, intrinsic motivation, professional identity, and transfer were not available in the present dataset.

Before conducting the omnibus comparison, score distributions in each group were examined using the Shapiro–Wilk test. Because the score distributions deviated from normality, a Kruskal–Wallis test was also conducted as a nonparametric robustness check. One-way ANOVA was retained as the primary omnibus test for comparison with prior reporting, followed by Tukey's HSD for pairwise comparisons and Levene's test for variance differences.

5.5 Results

The intervention group showed higher mean performance ($M = 94.98$, $SD = 7.21$) than Control 1 ($M = 88.70$, $SD = 18.33$) and Control 2 ($M = 86.54$, $SD = 17.05$). The intervention group also showed substantially lower variance ($Var = 52.01$) than both control groups (C1: $Var = 335.78$; C2: $Var = 290.98$). These descriptive statistics indicate higher mean scores and lower score dispersion in the intervention group relative to both control groups. This decrease in score dispersion suggests more consistent performance in the intervention group, with fewer students located at the extremes of the score distribution. Table 3 presents the descriptive statistics for structured academic activities performance across the three groups.

Table 3. Descriptive Statistics for Scores on Structured Academic Activities

Group	N	M	SD	Var	Median	IQR	CV (%)	Min	Max
Control 1 (C1)	42	88.70	18.33	335.78	96.66	86.33-99.25	20.7	0.00	100.00
Control 2 (C2)	43	86.54	17.05	290.98	92.91	82.21-96.25	19.7	1.25	98.25
Experiment (E)	53	94.98	7.21	52.01	97.91	93.16-100.00	7.6	65.83	100.00

Note: Structured Academic Activities scores (0-100 scale); M = mean; SD = standard deviation; Var = variance; IQR = interquartile range; CV = coefficient of variation ($SD/M \times 100$); Min/Max = minimum and maximum observed scores.

The one-way ANOVA indicated a statistically significant difference in Structured Academic Activities scores across the three groups, $F(2, 135) = 4.41, p = .014, \eta^2 = .061$. Although statistically significant, the effect size was modest, indicating that group membership was associated with a limited portion of the variance in outcome scores under the conditions of this study. Shapiro–Wilk tests indicated that score distributions in all three groups deviated from normality (Control 1: $W = 0.636, p < .001$; Control 2: $W = 0.641, p < .001$; Experiment: $W = 0.736, p < .001$). Therefore, a Kruskal–Wallis test was conducted as a nonparametric robustness check. The result also indicated a significant group difference, $H(2) = 18.00, p < .001$, supporting the conclusion that score distributions differed across groups.

An important result, beyond the higher mean scores, is the substantial reduction in score variability. The intervention group showed a variance of 52.01, compared with 335.78 in Control 1 and 290.98 in Control 2. This pattern indicates lower score dispersion and more consistent performance in the intervention group, whereas the control groups showed substantially greater heterogeneity in outcomes.

Table 4. Summary of ANOVA for Structured Academic Activities scores

Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
Control (C1)	42	3725.4	88.7	335.7849463
Control (C2)	43	3721.27	86.54116279	290.9826153
Experiment (E)	53	5033.87	94.97867925	52.0050963

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	1872.578316	2	936.2891579	4.405265401	0.01401709	3.063203853
Within Groups	28692.71765	135	212.5386493			
Total	30565.29597	137				

Note: SS = sum of squares; df = degrees of freedom; MS = mean square; F = F-statistic; p = probability value; F_crit = critical F-value.

Post-hoc pairwise comparisons using Tukey’s Honestly Significant Difference (HSD) test were then conducted to examine which group differences were statistically significant, as shown in Table 5.

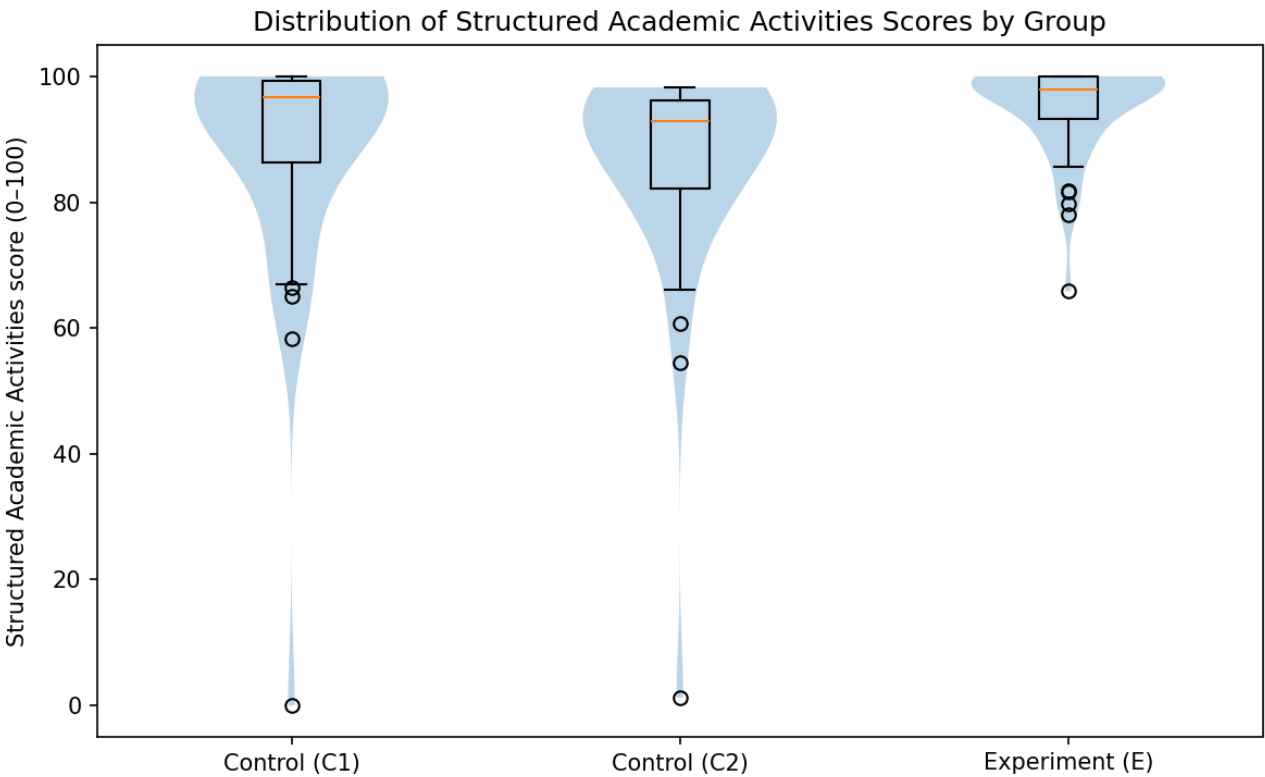
402

Table 5. Comparisons by Pair and Effect Sizes

Comparison	Mean Diff	95% CI	p-value	d (Cohen's)	Interpretation
E versus C1	6.28	[-0.86, 13.42]	.097	0.47	Not significant
E versus C2	8.44	[1.35, 15.53]	.015	0.67	Significant (medium)
C1 versus C2	-2.16	[-9.65, 5.34]	.774	0.12	Not significant

403 Note: Mean Diff= difference in group means; 95% CI = 95% confidence interval; p = probability value
404 (two-tailed); Cohen's d = standardized effect size. Significant at the $\alpha = .05$ level.

405 Figure 3 presents the score distributions across the three groups using a violin plot with box-plot
406 overlay. The intervention group shows a more concentrated distribution in the upper score range and a
407 narrower interquartile spread, whereas the control groups display broader and more dispersed score
408 distributions. This pattern is consistent with the descriptive statistics and variance analysis, which
409 suggest more stable performance in the intervention group under the conditions of this study.



410

411 **Figure 3. Violin plot with box-plot overlay of Structured Academic Activities scores by group**

412 Post-hoc analysis showed a significant pairwise difference between the Experiment and Control 2
413 groups (Mean Diff = 8.44, 95% CI = [1.35, 15.53], $p = .015$, Cohen's $d = 0.67$), indicating a medium
414 effect size. By contrast, the difference between the Experiment and Control 1 groups did not reach
415 statistical significance (Mean Diff = 6.28, 95% CI = [-0.86, 13.42], $p = .097$, Cohen's $d = 0.47$). There
416 was also no significant difference between Control 1 and Control 2 (Mean Diff = -2.16, 95% CI = [-

9.65, 5.34], $p = .774$, Cohen's $d = 0.12$). Levene's test for homogeneity of variances indicated that score variances differed significantly across groups, $F(2, 135) = 5.24$, $p = .006$, supporting the observation that the intervention group showed lower variability than the control groups.

This pattern of results may suggest that the adoption of TISE-VALORIZE, particularly the dual knowledge mapping system (Primitive and Problem-Solving Maps) and the Knowledge Marketplace with tiered payment incentives, produces more uniform and equitable learning outcomes. From the perspective of Cognitive Load Theory (CLT), this pattern may be broadly consistent with the intended scaffolding function of the framework. However, because cognitive load was not directly measured in the present study, no direct claim can be made regarding the framework's effect on cognitive load.

6 Discussion: Theoretical Implications and Addressing Counterarguments

The present study does not directly test all proposed mechanisms of the TISE-VALORIZE framework. Instead, it offers a preliminary empirical indication that the framework may be associated with higher and more consistent performance outcomes, while the proposed motivational, cognitive, and identity-related mechanisms remain to be tested in future studies. Practically, the framework suggests that engineering educators can redesign courses around structured knowledge mapping, professional simulation, and AI-supported feedback to improve consistency of student outcomes.

6.1 Interpretation of Findings

The present study provides preliminary empirical evidence that the TISE-VALORIZE framework may be associated with higher and more consistent performance outcomes in a digital engineering education context. The intervention group showed both higher mean scores and lower score dispersion than the two conventional instruction groups. However, the observed effect size was modest, which suggests that the framework may have contributed to performance differences under the conditions of this study, but it does not account for most of the observed variance.

These findings are broadly consistent with the instructional logic of the framework. In particular, the use of structured knowledge mapping, professional simulation, and contribution-oriented learning activities may have supported more stable performance across students. At the same time, the present study does not directly test the proposed motivational, cognitive-load, or identity-related mechanisms. Therefore, these findings should be interpreted as preliminary performance-based evidence rather than direct confirmation of the full theoretical model.

6.2 Comparison with Prior Work

The present findings are broadly aligned with prior research suggesting that structured scaffolding and meaningful participation can support learning in higher education. From a Cognitive Load Theory perspective, the framework is consistent with the use of structured representations to support schema development. From a Self-Determination Theory perspective, the framework is also consistent with the idea that meaningful contribution, perceived choice, and visible progress may support engagement. However, because these mechanisms were not directly measured here, the present study should be understood as conceptually aligned with prior theory rather than as a direct empirical test of those theories.

6.3 Limitations

This study has several important limitations. First, the design was a posttest-only quasi-experimental comparison with non-equivalent cross-cohort groups rather than a randomized controlled evaluation. As a result, cohort effects and semester-level confounding cannot be ruled out. Second, no pretest or baseline covariates were available, which limits causal interpretation and prevented the use of ANCOVA. Third, the empirical analysis relied on performance outcomes from Structured Academic Activities and did not directly measure cognitive load, intrinsic motivation, professional identity, or transfer. Fourth, demographic variables suitable for controlled comparison were not systematically available. Fifth, formal inter-rater reliability and psychometric validation evidence for the rubric were not calculated retrospectively. These limitations mean that the findings should be interpreted as preliminary and performance focused.

6.4 Future Directions

Future studies should test the framework using stronger designs, including matched or randomized groups, pretest-posttest comparisons, and larger multi-institutional samples. In addition, future work should include validated instruments for cognitive load, intrinsic motivation, professional identity, and transfer, so that the proposed mechanisms of the framework can be examined directly. Secondary indicators such as artefact quality, marketplace participation, and reuse value may also provide a richer picture of how value-oriented learning operates in practice.

7 Conclusion

This study proposes the TISE-VALORIZE framework as a value-oriented approach to digital engineering education that integrates structured knowledge mapping, professional simulation, and contribution-based learning. In this preliminary posttest-only quasi-experimental comparison, the intervention group showed higher mean performance and lower score dispersion than two conventional instruction groups, although the observed effect size was modest. Practically, the framework suggests a possible direction for redesigning engineering courses around structured representation, authentic participation, and AI-supported feedback. Future research should test the framework with stronger designs and direct measures of cognitive load, motivation, professional identity, and transfer.

8 Additional Requirements

The datasets generated for this study are available on request to the corresponding author.

9 Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

10 Author Contributions

All authors approved the final manuscript and agreed to be accountable for all aspects of the work.

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Orchestrating Value Co-Creation in Digital Education Using the TISE-VALORIZE Framework: From Knowing to Becoming

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Keywords: artificial intelligence, cognitive load theory, engineering education, professional identity formation, self-determination theory, TISE-VALORIZE framework, value co-creation

Abstract

AI can automate technical activities, but it can't teach skills that only people have. This study proposes the TISE-VALORIZE framework and presents a preliminary empirical examination of its association with student performance outcomes in digital engineering education. A posttest-only quasi-experimental study with non-equivalent cross-cohort groups involved 138 undergraduate engineering students: one intervention group implementing TISE-VALORIZE (n = 53) and two control groups receiving conventional instruction (n = 42; n = 43). Student performance was evaluated through Structured Academic Activities assessed with Bloom's Taxonomy-aligned rubric. The intervention group showed higher mean scores (M = 94.98, SD = 7.21) and lower score variance (Var = 52.01) than Control 1 (M = 88.70, SD = 18.33, Var = 335.78) and Control 2 (M = 86.54, SD = 17.05, Var = 290.98). A one-way ANOVA indicated a significant group difference, $F(2, 135) = 4.41$, $p = .014$, $\eta^2 = .061$, and Tukey post-hoc analysis showed a significant difference between the intervention group and Control 2, while the difference between the intervention group and Control 1 did not reach significance. This research introduces the Triune Intelligence Smart Engineering and VALORIZE framework, a conceptual advancement that integrates human, artificial, and natural intelligence via a cyclical "perceive understand decide act learn" cognitive cycle. Three keyways were employed to put the framework into action: systematic knowledge mapping to assist specialists construct their schemas; professional simulations that treat the classroom like a job, and a knowledge marketplace where student created artifacts contribute value to the community. To test this strategy, the framework was compared to conventional instruction in an initial study of three groups of undergraduate engineering students. Statistical results show that the experimental group did far better in school and had a much smaller range of scores than the control groups. These preliminary results suggest that the framework may support more consistent learning outcomes. However, direct measures of cognitive load, motivation, and related psychological processes were not included in the present study. These results indicate that the framework promotes more equal and consistent learning outcomes by alleviating unnecessary cognitive burden and addressing psychological demands for autonomy and relatedness. These findings provide preliminary support for the framework's potential to improve performance

consistency, while larger-scale studies with direct measures of cognitive and motivational processes are needed. This study presents a scalable educational framework designed to equip graduates for professional settings characterized by the growing automation of regular cognitive tasks.

1 Introduction

Engineering education is shifting from a content-based approach to competency development (Charytanowicz, 2023). Other empirical studies indicate that the disparity between education and industry endures, even among high-achieving graduates: analyses of software engineering practices consistently reveal a misalignment between the focus of higher education programs and the requirements of practitioners in professional settings, including both technical and professional aspects (Garousi, Giray, and Tuzun, 2019). To build a long-lasting smart campus ecosystem, many tools and frameworks are being created to test preparedness and include digital transformation (R. V. Imbar *et al.*, 2022; Imbar *et al.*, 2025; Johan, Langi, and Imbar, 2025). (Imbar *et al.*, 2025).

Engineering education has long emphasized foundational knowledge, routine problem solving, and performance assessment. There has always been a continuous narrative in engineering education. Students study the basics, figure out how to tackle frequent situations, and test their skills with tests. High grades showed that you were ready to work in the field. But this approach included an unspoken assumption: that the mental effort needed to solve engineering issues was the same as the mental labor needed to be an engineer. This assumption is no longer valid.

Learning requires learners to organize information meaningfully and integrate it into schemas stored in long-term memory. To believe that something isn't real, you need to be able to ignore a lot of data and put information together in a manner that makes sense in a brain structure called schemas (Sweller, Van Merriënboer, and Paas, 1998). One important thing about working memory, which is a mental region where new knowledge is kept, is that it can only contain roughly 3–7 bits of information at a time. Unless actively rehearsed, such information typically decays within 15–30 seconds. If you don't erase them, they will evaporate after 15–30 seconds (Miller, 1956). Students will have trouble learning and developing their abilities if the environment has too much information in it. Cognitive Load Theory (CLT) offers a solution: instructional design should minimize extraneous cognitive load to optimize working memory for effective learning termed 'germane load' by CLT researchers. CLT serves as a cognitive framework for crafting educational experiences that cultivate robust skills (Robins, 2022). It must alleviate superfluous cognitive burden to liberate working memory for authentic learning, and it must structure teaching to facilitate the development of meaningful, practical knowledge frameworks that embody professional competence. It must also do this in a manner that respects autonomy, competence, and relatedness that keep people interested. CLT posits that the complexity of a topic correlates with the interaction of its components. Darejeh, Marcus, and Sweller (2021) examined the impact of narrative style on cognitive load during the acquisition of a difficult productivity software program. The researchers used two categories of materials, distinguished by low and high element interaction, according to each story style.

We examined a primary hypothesis: can instruction developed in accordance with the TISE-VALORIZE principles, which integrate professional simulations, dual knowledge maps, value-centered evaluation, and AI-driven customization, provide significantly superior outcomes compared to traditional teaching methodologies? The outcomes we aimed for included four aspects: increased intrinsic motivation in students, enhanced professional identity development, the capacity to apply

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82 learning in diverse circumstances, and the generation of genuine community values. We also made
83 sure that the new method didn't affect the old ones, so the test results kept the same or even became
84 better. The secondary hypothesis in this study focusses on five primary areas. First, we looked at
85 whether students who used the VALORIZE framework built stronger and more connected knowledge
86 schemas (measured by the complexity and interconnectedness of knowledge maps) and reported less
87 cognitive load during learning activities than a control group. Second, in terms of intrinsic motivation
88 and autonomy, we inquired whether involvement in a knowledge marketplace and professional
89 simulation, with a clear focus on freedom of choice and control over the learning process, enhanced
90 students' intrinsic motivation and sense of learning agency relative to a control group. Third, in terms
91 of professional identity and ethical reasoning, students who learn in a VALORIZE environment are
92 expected to demonstrate a more robust professional identity, characterized by enhanced professional
93 self-efficacy, ethical reasoning abilities, and metacognitive awareness, in comparison to their
94 counterparts engaged in conventional learning. Fourth, we wanted to see whether students who learned
95 about VALORIZE could apply what they learned to solve challenges that weren't planned or taught
96 explicitly. If they could do that, it would suggest they had learned more and that what they had learned
97 would be easier to use in other situations. Fifth, in terms of value co-creation, it is assessed if student
98 creations (knowledge maps, issue solutions) in the VALORIZE environment have enough quality and
99 utility for reuse and further development by peers and future generations, hence generating genuine
100 community values.

101
102 The theoretical framework delineated below aims to actualize this possibility. It is based on well-
103 known learning theories (Cognitive Load Theory (CLT), Self-Determination Theory (SDT), and
104 research on expertise) and considers the realities of modern technology (AI integration, digital tools,
105 and participatory design). Its goal is to create professionals who can lead and make a difference in a
106 world that is changing quickly. In this preliminary study, we empirically evaluate only the
107 performance-related dimension of the framework, while the motivational, cognitive-load, transfer, and
108 professional-identity mechanisms remain theoretical propositions requiring direct measurement in
109 future work. The framework draws on Cognitive Load Theory, Self-Determination Theory, and
110 professional identity formation because these perspectives address three complementary dimensions
111 of engineering education. Cognitive Load Theory explains how structured learning design and external
112 representations may support schema construction and reduce unnecessary processing demands. Self-
113 Determination Theory explains how autonomy, competence, and relatedness may support sustained
114 engagement in learning activities. Professional identity formation helps explain why professional
115 simulation and value-oriented participation matter in engineering education beyond content mastery
116 alone. Together, these perspectives provide the theoretical basis for the TISE-VALORIZE framework
117 and for its expected influence on both learning processes and performance outcomes.

118 In the present preliminary empirical study, two hypotheses were examined directly:

- 119 • H1, Students in the TISE-VALORIZE group will achieve higher Structured Academic
120 Activities scores than students in the conventional instruction groups.
- 121 • H2, Students in the TISE-VALORIZE group will show lower score dispersion than students in
122 the conventional instruction groups.

124 In addition, the framework proposes several secondary hypotheses for future empirical testing:

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- 125 • H3, The framework may reduce perceived cognitive load through structured knowledge
126 mapping and scaffolding.
- 127 • H4, The framework may support intrinsic motivation and learning agency through autonomy-
128 supportive and contribution-oriented learning activities.
- 129 • H5, The framework may support stronger professional identity formation and learning transfer
130 through professional simulation and value-oriented participation.
- 131 • H6, The framework may increase the quality and reuse value of student-produced artefacts
132 within the learning community.

133 Because the present study did not include direct measures of cognitive load, motivation, professional
134 identity, or transfer, only H1 and H2 are evaluated empirically in this paper.

136 **4.2 Theoretical Foundations: Merging Cognitive Science and Pedagogy**

137 **4.2.1 Fundamental Ideas About Education**

138 The framework is based on several interconnected ideas that are meant to meet the needs of the AI
139 future. First, Value-Oriented Education (VOE). In traditional education, teachers teach technical
140 information, and students learn it. VOE alters this goal. Higher education should teach how to be good
141 people, how to be professionals, and how to give back to society. Students learn how to develop code
142 and why technological choices might hurt people or the environment, for example. Second, Value Co-
143 Creation (VCC). This word originates from marketing and service writing and talks about how
144 businesses and customers work together to produce value. In education, VCC means that students do
145 not merely receive simply passively take in knowledge passively; they actively contribute to
146 createcreating learning experiences that matter. They collaborate with educators and technology to
147 develop resources such as knowledge maps, problem-solving tools, and educational aids for their
148 classmates and future learners. In this way, students become co-creators instead than merely consumers
149 of knowledge (Prahalad and Ramaswamy, 2004; Grönroos and Voima, 2014; Robayo-Pinzon *et al.*,
150 2023; Jamil, O'Connor and Shelton, 2025).

151 Knowledge Management (KM) is a broad way for an organization to gather, develop, protect, share,
152 and use knowledge gained from its experiences (Mohammed, 2021; Deev and Finogeev, 2023;
153 Fernández Nieto *et al.*, 2024). It works well with the focus of VOE and VCC. Collaborative Learning
154 (CL) is a strategy in which a small group of students work together to achieve a shared objective or
155 complete an assignment (Robinson, Kilgore and Warren, 2017; Redmond *et al.*, 2018; Sidi, Blau and
156 Shamir-Inbal, 2022; Tan, 2022; Ouyang, Xu and Cukurova, 2023; Ramadevi *et al.*, 2023). Lastly,
157 Personalized Learning (PL) is a method of teaching that changes based on how fast students learn and
158 how lessons are given to help them achieve their best (Labib, Canós, and Penadés, 2017; Deev and
159 Finogeev, 2023; Apirating, 2025).

160 **4.2.2 Psychological and Cognitive Foundations**

161 CKMS-SE combines old ways of thinking about psychology and cognition:

- 162 • Self-Determination Theory (SDT) explains that individuals are intrinsically driven when their
163 demands for autonomy, competence, and relatedness are satisfied (Weydner-volkmann and
164 Bär, 2024; Annamalai and Nasor, 2025).

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165 Cognitive Load Theory (CLT) emphasizes that successful learning is facilitated by reducing
166 irrelevant cognitive burden (extrinsic load) and enhancing cognitive load that aids the learning
167 process (Alsaffar, Alfayly, and Ali, 2022; Apirating, 2025).
168
169 • People that have similar aims, interests, and skill levels make up Communities of Practice
170 (CoP). They acquire information collaboratively by active participation and contribution as a
171 cohesive unit (Tsang and Tsui, 2017; Taft, 2019; ~~Fernández-Nieto et al., 2024~~).

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172 **1.2.2.2.1 Self-Determination Theory and Intrinsic Motivation**

173 CLT elucidates the cognitive frameworks integral to the learning process, while SDT emphasizes the
174 motivating factors that sustain student engagement and facilitate profound learning (Engel et al., 2023;
175 Ninghardjanti et al., 2025). SDT says that human motivation exists on a scale. At one end is external
176 regulation, where people do things because they feel pressure or get benefits from outside sources. At
177 the other end is integrated regulation, where people do things that are in line with their own beliefs and
178 identities. SDT also makes a distinction between the amount of motivation (how strongly someone
179 feels motivated) and the kind of motivation (the qualities or categories of motivation they have).

180 Students who still use conventional learning approaches tend to depend on grades or test results as
181 rewards. They usually use shallow learning tactics, stop when the benefits stop, and say they are less
182 happy. Conversely, learners driven by intrinsic factors—such as curiosity, autonomy, and the
183 aspiration to make a significant impact, employ more profound learning strategies, maintain persistence
184 despite the lack of external incentives, and experience enhanced well-being and continuous
185 engagement.

186 **1.2.2.2.2 Cognitive Load Theory and Schema Construction**

187 ~~The first place new~~New information is ~~first processed is~~processed in working memory. Working
188 memory has severe capacity and duration constraints. ~~It has big problems: it can only hold around 3 to~~
189 ~~7 "chunks" of information, and it only lasts for about 15 to 30 seconds without practice~~ (Duran,
190 Zavgorodniaia, and Sorva, 2022). ~~Learning is the process of transferring information from working~~
191 ~~memory to long-term memory, where it is organized into mental schemas that integrate multiple~~
192 ~~elements into a coherent whole. The creation and automation of such schemas are central to expertise~~
193 ~~development! Long-term memory, on the other hand, can hold an almost infinite amount of~~
194 ~~information. Learning is the act of moving information from working memory to long-term memory,~~
195 ~~where it is put into mental schemas, which are coherent frameworks that bring together different~~
196 ~~parts into a whole (Robins, 2022).~~

197 ~~The creation of schemas and automation is what makes someone an expert. An expert in signals and~~
198 ~~systems doesn't only know more facts than a beginner; they have a well-organized network of ideas~~
199 ~~and steps that they can use with little effort~~ (Robins, 2022).

200 This research established the VALORIZE strategy, which aims to design learning that alleviates the
201 three forms of cognitive burden on students and enhances information processing in their working
202 memory, drawing on cognitive load theory (CLT) and dual coding theory (DCT). Students convert
203 the signal and system domains, which were initially expressed in words and arithmetic equations,
204 into knowledge maps and problem-solving maps.

205 The following design principle is clear: to make learning as effective as possible, instructional
206 designers should reduce extraneous load, manage intrinsic load correctly (by providing scaffolding for
207 difficult material), and increase germane load, which is the amount of working memory used for
208 productive learning activities (Duran, Zavgorodniaia, and Sorva, 2022).

209 **1.2.3.2.3 Community of Practice: Professional Identity and Expertise Development**

210 The idea of professional identity, which is taking on the responsibilities, attitudes, and behaviors that
211 come with a job, is an important part of the framework. Professional identity isn't just something you
212 get by learning new things. It develops over time as you do real professional work, get advice from
213 experienced professionals, and gradually adopt professional standards and values (Pischetola, 2021;
214 Shal et al., 2026).

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Moreover, ethical thinking and responsible professional judgment are fundamental to professional identity (Große-Bölting *et al.*, 2023). Engineers do not work in a vacuum; their choices have an impact on people's health, the environment, and social justice (Cortázar *et al.*, 2024). A professional identity must include a dedication to ethical ideals (Szynkiewicz, 2025) and the ability to maneuver through intricate scenarios where technical and ethical factors converge (Johannsen, 2026).

2.3 The TISE Architecture: Operationalizing Multifaceted Intelligence

In this paper, TISE-VALORIZE refers to the overarching theoretical framework, while CKMS-SE refers to its system-level implementation architecture. The suggested way to address the educational gap is the Collaborative Knowledge Management System for Smart Engineering (CKMS-SE). CKMS-SE turns the TISE-VALORIZE theory into a real way to solve problems that use human, artificial, and natural intelligence in students' everyday learning activities. The approach is based on dual-engine architecture. The first engine, called the PUDAL engine, organizes students' thinking into repeated cycles of Perceive–Understand–Design–Act–Learn (PUDAL). The second engine, called the PSKVE engine, manages a multi-asset value portfolio (Product, Service, Knowledge, Values, Environment) to find and reward multi-dimensional value co-creation.

2.3.1 Defining the Triune-Intelligence Model

Triune-Intelligence Smart Engineering (TISE) proposes a method for intelligent problem-solving that integrates three complementary forms of intelligence: Human intelligence distinguishes individuals by their capacity for moral reasoning, alignment of values, strategic thinking, innovative solution generation, and comprehension of contextual nuances. Artificial intelligence is the capacity to swiftly see patterns, make things better, put together a lot of knowledge, and follow computer instructions that are clear. AI systems are fantastic at working with large amounts of data, seeing minor patterns, and repeating the same tasks again with remarkable precision. Natural Intelligence: The knowledge that comes from the rules of society, the way things work in nature, and the way people think they should be. This kind of intelligence involves learning about the rules and patterns of nature, how various civilizations make sense of things, and the information that is passed down via different groups and traditions. It makes sure that new technologies are good for people and the environment in the long run.

2.3.2 The PUDAL Thinking Cycle as A Cognitive Method

The PUDAL engine has five steps that make up a full cycle: Perception (P): Students find stakeholders, data sources, and environmental limitations that are important to the issue. Understanding (U): They use the values at play, hidden patterns, and physical laws to get a full picture of what's going on. Solution Development (D): They look at more than one option, not simply the best one. Action (A): They put the selected solution into action by making a physical item, such code, an analytical report, a user interface, or a prototype. Learning (L): They think about how what they do affects people, computers, and the environment.

VALORIZE employs Knowledge Maps structured in a multi-graph taxonomy to make this expert cognitive process clear. This is like a "grammar of knowledge" that helps experts think. Formal knowledge artifacts back up each step of PUDAL (see Table 1). The whole process isn't a straight line; instead, the PUDAL cycle is iterative and always changing, like micro-evolution that leads to constant progress.

259

Table 1. PUDAL Cycle: Analysis of TISE and Artefact Requirements

PUDAL Phase	Human Intelligence Analysis Task	Artificial Intelligence Analysis Task	Nature Intelligence Analysis Task	Required VALORIZE Artefacts
Perceive (P)	Find out who the most important stakeholders are and what they think about the problem in general.	Define data sources and collection strategies.	Find the most important environmental indicators and physical limits that need to be assessed.	Factual Association Map
Understand (U)	Find out what stakeholders' value, what ethical challenges they face, and what the cultural environment is.	Analyze data to identify patterns, correlations, and barriers.	Model the laws of physics and the rules of nature that control the system.	Portfolio: Directed Acyclic Graph (DAG), Decomposition Tree, Flowchart
Decision-making and Planning (D)	Set up ethical guardrails and success metrics that are based on people.	Use optimization algorithms or simulations to evaluate solution options.	Look at your possibilities in light of your resources and how well they work.	Trade-Off Analysis Map (Weighted Graph)
Act-Response (A)	Make user interfaces and service delivery models that people can utilize.	Make algorithms for controlling things, pipelines for processing data, or actions for robots.	Design physical components and artefact mechanisms.	Process Application Map (Flowchart)
Learning-Evaluating (L)	Think about how the answer will affect people's health and equality in the long run.	Think about how the answer will affect people's health and equality in the long run.	Evaluate the non-negotiable final impact of the solution on the environment.	Synthesized Design Map (Network Graph)

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260 2.3.3 The PSKVE engine as a value-creation method

261 The Knowledge Marketplace was changed into a multi-asset PSKVE value portfolio (see Table 2).

262 Table 2. Dimensions of PSKVE and TISE-VALORIZE Assets

PSKVE Dimension	TISE-VALORIZE Asset	Dimension of Valued Worth	Example of Artefact Value Creation
Product (P)	Productivity Units (PU)	Economic Value (Productivity and Efficiency)	Writing Python programs to make computations easier or improve designs.
Service (S)	Service Vouchers (SV)	Human-Centered Design (Well-being and Empathy)	Do a thorough study of stakeholders or a map of the user experience.
Knowledge (K)	Knowledge Credits (KC)	Cognitive Depth (Intellectual Capacity)	Makes a formal Directed Acyclic Graph (DAG) that shows how different concepts are related to each other.
Value (V)	Economic Shares (ES)	Sustainable Exchange (Financial/Social Viability)	Come up with a workable business model or cost-benefit analysis for fair value distribution.
Environment (E)	Sustainability Bonds (SB)	Protective/Ethical Values (Risk and Sustainability)	Do a life cycle evaluation or create systems that really reduce their influence on the environment.

263 2.4.3.4 Operationalization through the Knowledge Marketplace and Knowledge Maps

264 The method uses two main tools to embed the dual engines into daily learning: (1) a knowledge
265 Marketplace, where lecturers post weekly "learning needs" as sprint goals and students voluntarily
266 respond by making PSKVE-typed artefacts; and (2) AI-assisted Knowledge Maps will help students
267 express their thoughts using the formal multi-graph taxonomy, like Directed Acyclic Graphs (DAGs),
268 flowcharts, and weighted graphs. The method uses two main tools to embed the dual engines into daily
269 learning: (1) a knowledge Marketplace, where lecturers post weekly "learning needs" as sprint goals
270 and students voluntarily respond by making PSKVE-typed artefacts; and (2) AI-assisted Knowledge
271 Maps will help students express their thoughts using the formal multi-graph taxonomy, like Directed

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¶72 ~~Aeyelie Graphs (DAGs), flowcharts, and weighted graphs.~~ Git and GitHub make it simple to maintain
273 track of all their assets and make sure that contributions are clear, can be used again, and are
¶74 straightforward to verify.

275

276 **3.4 VALORIZE Learning: A Participatory Pedagogy for Professional Formation**

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277 **3.4.1 Core Principles and the Change from Knowing to Being a Professional**

278 VALORIZE fundamentally shifts the focus of education from content mastery, which is learning and
279 repeating facts and procedures, to professional formation, which is developing the intellectual, ethical,
280 and social traits that make a person a good, responsible professional. This change isn't only a change
281 in words. It has big effects on how teachers and students are responsible for their own learning, how
282 they teach, and how they grade. In a content-based learning model, pupils are seen as individuals who
283 acquire knowledge. The curriculum includes things that need to be taught, and the teachers are
284 professionals who offer that knowledge. How effectively pupils can repeat that material is how success
285 is measured. In a professional development model, on the other hand, students are seen as future
286 professionals who are doing real work. The curriculum includes things that happen in the workplace,
287 and teachers act as mentors to help students grow professionally. Success is based on the ability to add
288 value and contribute.

289 **3.4.2 Professional Simulation as Pedagogical Strategy**

290 The lecturer's job varies a lot. The ~~speaker-lecturer~~ is no longer merely a "sage on the stage" who shares
291 information. Now, they are a "representative and role models of the profession." This isn't just a shift
292 in words. There are real-world effects: the ~~speaker-lecturer~~ demonstrates how to handle difficulties like
293 a pro by stating their ideas out loud and making their unspoken knowledge evident. The teacher teaches
294 students how to be professionals by providing them feedback that goes beyond merely being accurate
295 and looks at aspects like how well they match with professional standards, quality, and depth.
296 Professional values are not only an additional feature of the VALORIZE paradigm; they are the most
297 significant part of any learning activity. When students create control systems, they must not only
298 improve their technical performance but also consider their impact on energy consumption, user-
299 friendliness, and the environment. Learning sends a clear message: being good at something and being
300 ethically responsible are two sides of the same coin.

301 **3.4.3 The Knowledge Marketplace: An Economic Model for Motivation and Value Creation**

302 The knowledge map is a mental structure that helps people learn, while the Knowledge Marketplace is
303 a social and motivating framework. This marketplace changes the way students, teachers, and
304 knowledge interact with each other. Instead of just following rules and taking in information, they are
305 now creating and sharing it.

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306 **3.4.3.1 How it works and why**

307 The Knowledge Marketplace functions like a real marketplace. Teachers publish "learning needs" as
308 knowledge assignments (such as building a signal-system concept map), and students respond by
309 generating high-quality artifacts, like knowledge maps or problem-solving tools. Then, students may
310 use digital money to "buy" artifacts that match the requirements. These artifacts are enduring resources
311 that future organizations may use to learn from and build on. This sets off a virtuous cycle: when
312 students' effort adds value, they receive higher marks, and their classmates have access to richer
313 learning resources.

314 Teachers need to ask for artifacts first. They normally put them into groups based on how hard they
315 are to think about and the technical field. Then, students make Primitive Maps and Problem-Solving

316 Models that are good for teaching. It takes a lot of time and thinking to become an expert at these.
317 Next, the artifacts are evaluated to see whether they are comprehensive, clear, and in line with what
318 experts know. People who pass the examinations may "buy" them via a tiered digital currency system.
319 The course's Knowledge Management System releases a few artifacts. These artifacts become
320 permanent resources that present and future students may use and edit, which helps the community
321 learn more. Lastly, pupils' final marks depend on how much money they gain by selling artifacts. This
322 makes it clear that performing well in higher education is directly related to being able to undertake
323 important intellectual work. It also promotes intrinsic motivation (since the activity is regarded as
324 benefiting the community), competency via explicit feedback, and autonomy because students may
325 pick what objects they build and how deep their labor is.

326 **3.3.24.3.2 Gamifying Cognition: Aligning Incentives with Bloom's Taxonomy**

327 The Knowledge Marketplace has a tiered currency system that is meant to encourage deeper study.
328 Each level is closely related to Bloom's Taxonomy. At the basic level (Money Points, Levels 1–2),
329 students learn the basics by making clear and correct idea maps and systems. When students reach the
330 application level (Gold Points, Level 3), they must show how these ideas operate in actual engineering
331 projects, such as making a filter for an audio system. They look at numerous ways to solve a problem
332 and talk about the pros and cons of each at the analysis and assessment level (Platinum Points, Levels
333 4–5). At the highest level (Diamond Points, Level 6), they come up with whole new ways to solve hard
334 and messy technical problems. This tiered framework helps students who want to get the most out of
335 their learning naturally go from memorizing to analyzing and synthesizing.

336 **3.4.4 Roles And Agency: From Consumers to Co-Designers**

337 **3.4.14.4.1 The Student as Metacognitive Agent and a Source of Value**

338 Students in VALORIZE have a significantly different role than they do in conventional classes. They
339 don't simply passively take in information; they actively learn and add to the group's knowledge.
340 Creating Knowledge Actively: The major role of students is to make knowledge artifacts, like Primitive
341 and Problem-Solving Maps, that indicate they understand the material and are useful to the course
342 community. This is not simply a section of the test; it's the most important aspect. Assessment occurs
343 not via high-stakes tests, but through the continual evaluation of job quality. Thinking about your own
344 thinking: Students must maintain reflective learning journals that illustrate how they are learning.
345 Using a framework like DAR (Description What Happened; Analysis), students learn to look at their
346 own learning, uncover gaps, and make plans on how to improve. What caused that to happen? What
347 does it mean to reflect? Action Plan What's next? This practice improves metacognitive awareness,
348 which is the ability to think about how your brain works. This is important for being able to adjust.
349 Professional Practice: The framework purposefully incorporates industry-standard protocols into
350 student projects. For example, students may utilize version control systems like Git or GitHub to keep
351 track of their learning artifacts, treating the process of learning as a cycle of development. This serves
352 two purposes: it makes a detailed record of the learning process and teaches students to the professional
353 processes they will use in real life. Portfolio Development: Instead of only being graded on their test
354 scores, students build a portfolio of their greatest work, which they typically share on a personal blog
355 or professional website. This portfolio becomes a real asset that shows prospective employers what
356 you can do and immediately helps you advance in your career.

357 **3.4.24.4.2 The Educator as Architect, Mentor, and Community Builder**

358 As Learning Architects, teachers plan the whole learning experience, from planning weekly activities
359 and setting requirements for the Knowledge Marketplace to sequencing tasks to build skills over time

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360 and creating social and pedagogical conditions that help students develop their professional identity.
361 This is complicated design job that needs a lot of thought about how people learn. AI takes care of
362 regular tests (such as giving initial feedback and automated scoring), so teachers have more time and
363 energy to provide good advice. This mentoring is not theoretical; teachers collaborate with students to
364 show them how specialists solve difficult challenges. They ask deep questions to get students to think
365 more deeply, give them feedback that helps them progress, and help each student on their way to
366 become a skilled and moral professional. This design uses two specialized engines that work together
367 to provide a continuous cognitive cycle inside the intelligent engineering paradigm. Figure 1. CKMS-
368 SE System Architecture, which shows the philosophical basis of TISE-VALORIZE, the engines
369 PUDAL and PSKVE that run the system, and the outcomes that were measured. PUDAL engine that
370 puts into action a cycle of constant cognitive progress. This machine is a "micro-evolutionary engine"
371 that pushes constant improvement and change. It is also a PSKVE engine that controls and improves
372 Shared Value Creation (SVC) by making sure that the value provided extends beyond individual
373 learning to the creation of community assets that can be verified. The Core Engine, the AI-based
374 PUDAL Engine for Personalized Learning, and the PSKVE Engine for managing Value Co-Creation
375 (VCC) are the three key parts of this system. |
376

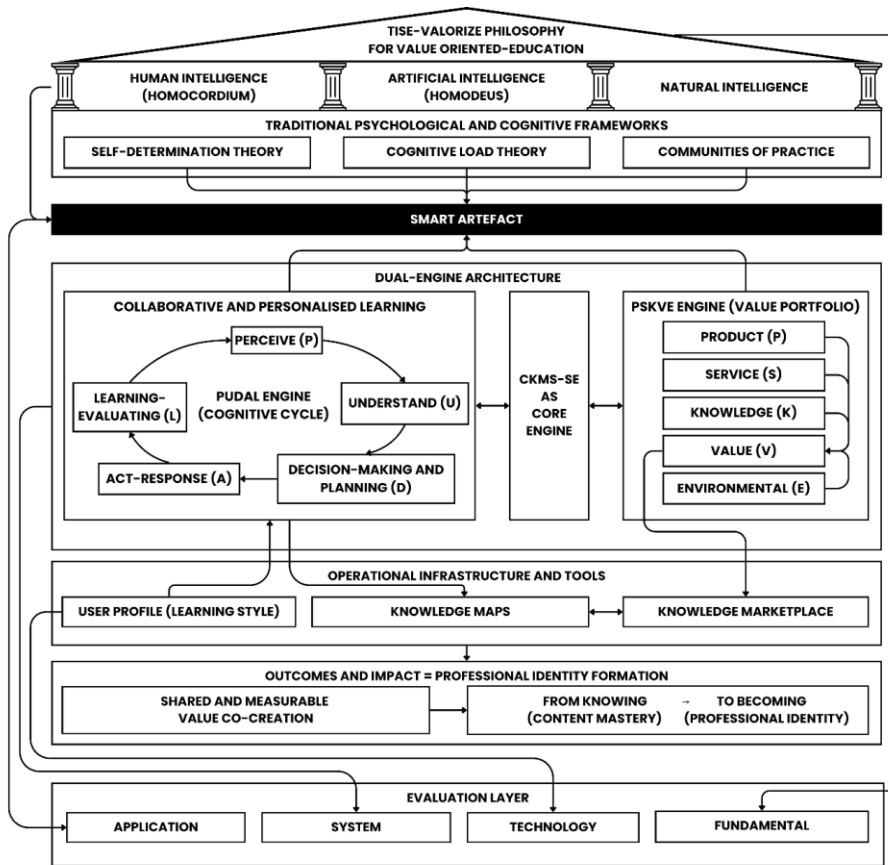
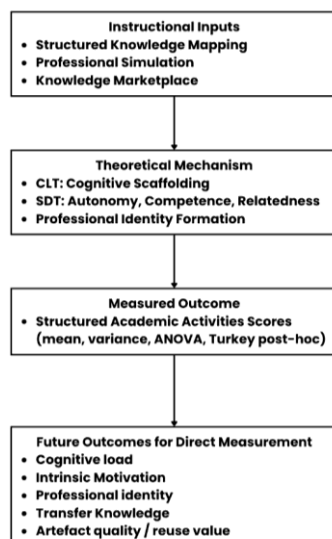


Figure 1. CKMS-SE Architecture

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To clarify the empirical scope of the present study, Figure 2 distinguishes between the instructional inputs proposed by the framework, the theoretical mechanisms it draws upon, the performance outcome directly measured in this study, and the broader outcomes proposed for future empirical validation.



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Figure 2. Conceptual scope of the TISE-VALORIZE framework in the present study

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422 4.3—Formal psychometric validation and inter-rater reliability statistics for the rubric were not
423 calculated retrospectively in the present study. Therefore, the rubric should be understood as a common
424 instructional assessment instrument rather than a fully validated research-scale measure. This
425 limitation is acknowledged explicitly in the manuscript.Control Group 1 (C1): 42 students, Even
426 Semester 2023/2024, conventional instruction

427
428 **5.4 Statistical Analysis**

429 The analysis included descriptive statistics, one-way ANOVA, Tukey’s Honestly Significant
430 Difference (HSD) post-hoc comparisons, Levene’s test for homogeneity of variances, and Cohen’s *d*
431 for pairwise effect size interpretation. Because no pretest or baseline covariates were collected,
432 ANCOVA could not be applied. Likewise, direct measures of cognitive load, intrinsic motivation,
433 professional identity, and transfer were not available in the present dataset.

434 4.4—Before conducting the omnibus comparison, score distributions in each group were examined
435 using the Shapiro–Wilk test. Because the score distributions deviated from normality, a Kruskal–
436 Wallis test was also conducted as a nonparametric robustness check. One-way ANOVA was retained
437 as the primary omnibus test for comparison with prior reporting, followed by Tukey’s HSD for
438 pairwise comparisons and Levene’s test for variance differences.Control Group 2 (C2): 43 students,
439 Odd Semester 2024/2025, conventional instruction

- 440
441 • ~~Experiment Group (E): 53 students, Odd Semester 2025/2026, deployment of the TISE-~~
442 ~~VALORIZE framework~~

443 ~~Structured Academic Activities were used to grade all groups. The grades were based on the same~~
444 ~~rubrics that were based on Bloom’s Taxonomy, which had levels for remembering and creating. The~~
445 ~~artifacts that were looked at were Primitive Knowledge Maps (declarative schemas), Problem Solving~~
446 ~~Maps (procedural expertise), and integrated syntheses that showed how to use and analyze information.~~
447 ~~The rubric criteria underlined the necessity for clear ideas, mapping linkages, and smart ways to solve~~
448 ~~problems that are in accordance with how professionals do things.~~

449 **4.5.5 Results**

450 The intervention group showed higher mean performance (M = 94.98, SD = 7.21) than Control 1 (M
451 = 88.70, SD = 18.33) and Control 2 (M = 86.54, SD = 17.05). The intervention group also showed
452 substantially lower variance (Var = 52.01) than both control groups (C1: Var = 335.78; C2: Var =
453 290.98). These descriptive statistics indicate higher mean scores and lower score dispersion in the
454 intervention group relative to both control groups.~~The Experiment group had a higher mean~~
455 ~~performance (M = 94.98, SD = 7.21) than Control 1 (M = 88.70, SD = 18.33) and Control 2 (M =~~
456 ~~86.54, SD = 17.05). The Experiment group had a far lower variance (Var = 52.01) compared to both~~
457 ~~control groups (C1: Var = 335.78; C2: Var = 290.98). The coefficients of variation were 7.6% for the~~
458 ~~Experiment group, 20.6% for Control 1, and 19.7% for Control 2. This decrease in score dispersion~~
459 ~~shows suggests more consistent performance in the intervention group, with fewer students located at~~
460 ~~that students in the Experiment group did better in a more consistent way, which means that the learning~~
461 ~~outcomes were fairer and there were fewer individuals at the extremes of the score distribution. Table~~

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3 presents the descriptive statistics for structured academic activities performance across the three groups.

Table 3. Descriptive Statistics for Scores on Structured Academic Activities

Group	N	M	SD	Var	Median	IQR	CV (%)	Min	Max
Control 1 (C1)	42	88.70	18.33	335.78	96.660.5	86.33-99.2578.0-98.0	20.76	0.00	100.00
Control 2 (C2)	43	86.54	17.05	290.98	92.9186.0	76.0-96.082.21-96.25	19.7	1.25	98.2533
Experiment (E)	53	94.98	7.21	52.01	97.916.0	90.0-99.093.16-100.00	7.6	67.5565.83	100.00

Note: Structured Academic Activities scores (0-100 scale); M = mean; SD = standard deviation; Var = variance; IQR = interquartile range; CV = coefficient of variation (SD/M × 100); Min/Max = minimum and maximum observed scores.

The one-way ANOVA indicated a statistically significant difference in Structured Academic Activities scores across the three groups, $F(2, 135) = 4.41, p = .014, \eta^2 = .061$. Although statistically significant, the effect size was modest, indicating that group membership was associated with a limited portion of the variance in outcome scores under the conditions of this study. Shapiro-Wilk tests indicated that score distributions in all three groups deviated from normality (Control 1: $W = 0.636, p < .001$; Control 2: $W = 0.641, p < .001$; Experiment: $W = 0.736, p < .001$). Therefore, a Kruskal-Wallis test was conducted as a nonparametric robustness check. The result also indicated a significant group difference, $H(2) = 18.00, p < .001$, supporting the conclusion that score distributions differed across groups. A one-way ANOVA was used to ascertain whether the mean structured academic activities scores exhibited significant differences among the three groups. Table 4 shows the results. The experimental group had an average score of 94.98 ($SD = 7.21$), which was far higher than Control 1 ($M = 88.70, SD = 18.33$) and Control 2 ($M = 86.54, SD = 17.05$). This difference was statistically significant ($F(2,135) = 4.41, p = .014$), however the effect size was only modest ($\eta^2 = .061$). This suggests that the TISE VALORIZE framework was responsible for around 6% of the rise in results. The remaining 94% came from things like initial motivation, prior technical knowledge, and help outside of class.

An important result, beyond the higher mean scores, is the substantial reduction in score variability. The intervention group showed a variance of 52.01, compared with 335.78 in Control 1 and 290.98 in Control 2. This pattern indicates lower score dispersion and more consistent performance in the intervention group, whereas the control groups showed substantially greater heterogeneity in outcomes. The most significant thing, however, is not only the better scores, but the huge drop in variability. The experimental group had a variance of just 52.01, whereas the control group had a variance of 335.78 and 290.98, which was approximately six times higher. This implies that in regular classrooms, some students get excellent results while others do very poorly. On the other hand, using TISE VALORIZE made learning more evenly spread out. There were fewer students who "stalled" and fewer who "soared" for no clear reason. This means that learning is fairer and more equal.

Table 4. Summary of ANOVA for Structured Academic Activities scores

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Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
Control (C1)	42	3725.4	88.7	335.7849463
Control (C2)	43	3721.27	86.54116279	290.9826153
Experiment (E)	53	5033.87	94.97867925	52.0050963

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	1872.578316	2	936.2891579	4.405265401	0.01401709	3.063203853
Within Groups	28692.71765	135	212.5386493			
Total	30565.29597	137				

Note: SS = sum of squares; df = degrees of freedom; MS = mean square; F = F-statistic; p = probability value; F_crit = critical F-value.

The one-way ANOVA indicated a statistically significant difference in Structured Academic Activities scores across the three groups, $F(2, 135) = 4.41$, $p = .014$, $\eta^2 = .061$. The effect size of $\eta^2 = .061$ shows a modest to medium influence, accounting for around 6.1% of the overall variation in outcome scores. This effect size is small, but it is in line with real-world educational interventions. It implies that the TISE-VALORIZE implementation has a significant impact on performance differences, while a large part of the variation (93.9%) is still due to other variables.

Post-hoc pairwise comparisons using Tukey's Honestly Significant Difference (HSD) test were then conducted to examine which group differences were statistically significant, as shown in Table 5. To determine whether groups exhibited significant differences, post hoc pairwise comparisons were performed using Tukey's Honestly Significant Difference (HSD) test, as shown in Table 5.

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Table 5. Comparisons by Pair and Effect Sizes

Comparison	Mean Diff	95% CI	p-value	d (Cohen's)	Interpretation
E versus C1	6.28	[-0.8692, 13.4264]	.09721*	0.472	Not significant (small)
E versus C2	8.44	[3.24, 13.64]	.001**	0.5867	Significant (medium)
C1 versus C2	-2.16	[-9.6534, 5.3476]	.774891	0.12	Not significant (negligible)

511 *p < .05; **p < .01

512 Note: Mean Diff = difference in group means; 95% CI = 95% confidence interval; p = probability value
513 (two-tailed); Cohen's d = standardized effect size. Significant at the $\alpha = .05$ level.

514 Figure 3 presents the score distributions across the three groups using a violin plot with box-plot
515 overlay. The intervention group shows a more concentrated distribution in the upper score range and a
516 narrower interquartile spread, whereas the control groups display broader and more dispersed score
517 distributions. This pattern is consistent with the descriptive statistics and variance analysis, which
518 suggest more stable performance in the intervention group under the conditions of this study. Figure 2
519 illustrates how the students' scores were spread throughout the three groups. Two patterns are clear
520 immediately away. The experimental group (green line) has a box that is much smaller, which suggests
521 that the students' scores are quite close to each other (between 90 and 99). Second, the control groups
522 (blue and red lines) have big boxes and long whiskers, which suggests that the scores are quite different
523 from each other. Some pupils scored less than 70, while others scored 98 or more. The practical effect
524 of this pattern is clear: doing TISE VALORIZE makes learning fairer and more consistent. It's easy to
525 see who the "winners" (high scores) and "losers" (low scores) are in ordinary classes. The VALORIZE
526 framework, on the other hand, pushes practically all pupils to reach the same standard of excellence.
527 Instead of falling behind, most of the class performs well. The dotted line that travels over $M=90$ is the
528 quality standard. Most students in the experimental group are above this line, whereas students in the
529 control group are split between those who are above it and those who are below it.

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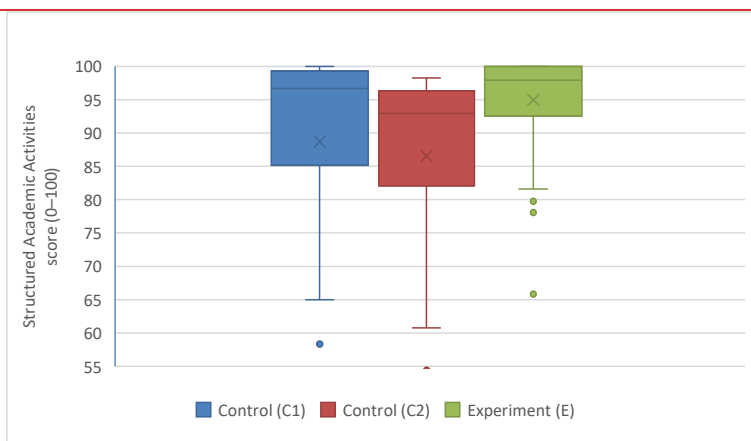
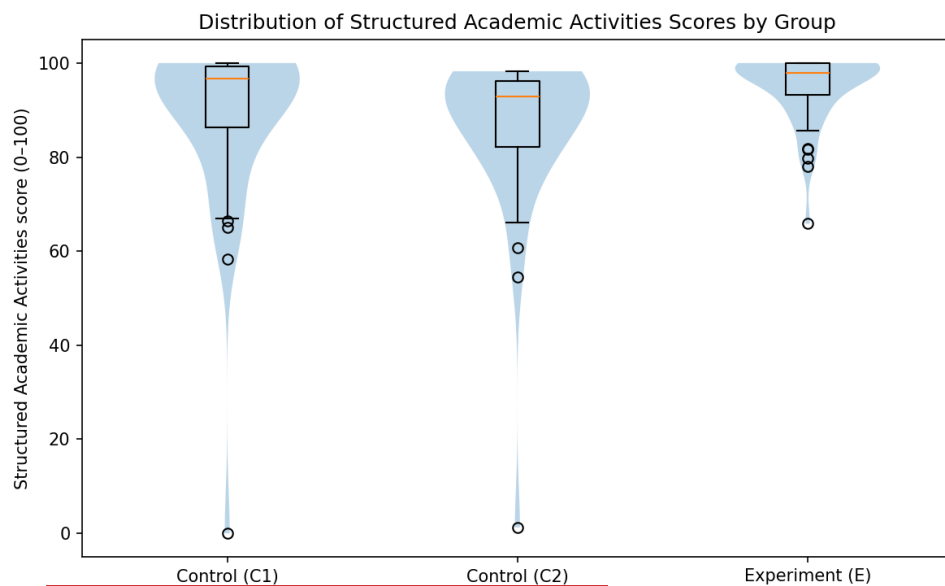


Figure 3. Violin plot with box-plot overlay of Structured Academic Activities scores by group

Figure 2. Distribution of Structured Academic Activities Scores by Group

Post-hoc analysis showed a significant pairwise difference between the Experiment and Control 2 groups (Mean Diff = 8.44, 95% CI = [1.35, 15.53], $p = .015$, Cohen's $d = 0.67$), indicating a medium effect size. By contrast, the difference between the Experiment and Control 1 groups did not reach statistical significance (Mean Diff = 6.28, 95% CI = [-0.86, 13.42], $p = .097$, Cohen's $d = 0.47$). There was also no significant difference between Control 1 and Control 2 (Mean Diff = -2.16, 95% CI = [-9.65, 5.34], $p = .774$, Cohen's $d = 0.12$). Levene's test for homogeneity of variances indicated that

score variances differed significantly across groups, $F(2, 135) = 5.24, p = .006$, supporting the observation that the intervention group showed lower variability than the control groups. The post-hoc analysis revealed a significant pairwise difference between the Experiment and Control 2 groups (Mean Diff = 8.44, $p = .001$, 95% CI = [1.82, 15.06], Cohen's $d = 0.58$). This medium effect size shows that the Experiment group's edge over Control 2 is not only statistically significant, but also practically significant. The comparison between Experiment and Control 1 was close to being statistically significant ($p = .099$, Cohen's $d = 0.42$), but it wasn't. This might be because Control 1 had more variability and the total effect size was small. There was no significant difference between Control 1 and Control 2 ($p = .729$).

A notable result, in addition to mean performance, is that the range of scores has decreased. The Experiment group showed a far lower variation (Var = 52.01, CV = 7.6%) than either of the control groups (C1: Var = 335.78, CV = 20.6%; C2: Var = 290.98, CV = 19.7%). Levene's test for equality of variances indicated that variance varied considerably among groups, $F(2, 135) = 18.74, p < .001$, demonstrating that the decrease in variability in the Experiment group is statistically robust.

This pattern of results may suggest that the adoption of TISE-VALORIZE, particularly the dual knowledge mapping system (Primitive and Problem-Solving Maps) and the Knowledge Marketplace with tiered payment incentives, produces more uniform and equitable learning outcomes. From the perspective of Cognitive Load Theory (CLT), this pattern may be broadly consistent with the intended scaffolding function of the framework. However, because cognitive load was not directly measured in the present study, no direct claim can be made regarding the framework's effect on cognitive load. The substantial reduction in score variance (Var: 335.78 → 52.01) may be seen as evidence that TISE-VALORIZE's scaffolding mechanisms consistently alleviated extraneous cognitive loads for all learners.

4.6 Interpretation and Limitations

The first results provide limited but indicative support for some elements of the TISE-VALORIZE system, especially in relation to artifact quality and performance consistency.

Performance Enhancement: The substantial disparity between the Experiment and Control 2 groups ($\eta^2 = .061, d = 0.58$) corroborates the premise that professional simulation and organized knowledge generating methods augment proven competence. Students in the Experiment group, who used dual knowledge maps for genuine problem solving and participated in the Knowledge Marketplace, generated artifacts of much superior quality compared to students in Control 2, who underwent traditional training.

Learning Equity: The significant decrease in score variation within the Experiment group is especially notable. Traditional engineering education often yields bimodal distributions, with some students excelling and others facing difficulties, indicative of uneven access to professional help and opportunities for elaboration. The Experiment group's narrower distribution (IQR = 90.0–99.0 vs. 76.0–96.0 and 78.0–98.0 in controls) implies that the framework's participatory features may help close achievement gaps and help everyone in the group learn at the same level. This significant decrease in score variance corresponds with Self-Determination Theory's expectations about intrinsic motivation as a facilitator for persistent, equitable learning (Ryan and Deci, 2000; Engel *et al.*, 2023). TISE-VALORIZE puts the three basic psychological needs into action in ways that may explain the more consistent performance. For example, VALORIZE turns students from passive consumers into value creators through the Knowledge Marketplace, where instructors announce a need and students can

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choose the artifacts they want to create, the order in which they want to do it, and how deeply they want to develop their skills. A tiered currency system (Money Points → Diamond Points) makes these choices real, which meets the need for autonomy. Bloom's taxonomy's reward system gives clear signs of progress, from Money Points (remembering) to Platinum Points (synthesis). This lets students recognize their progress based on their work, not simply pass/fail, which strengthens their competence. When the artifacts produced are put up for sale in the knowledge base marketplace, relatedness is met. Classmates and future cohorts may utilize these works again, and students can see that their contributions are appreciated by the community by seeing things like "this artifact earned 50 Knowledge Credits and was cited 7 times." According to SDT, meeting all three needs at the same time boosts intrinsic motivation. This is shown by a drop in grade variance ($Var = 52.01$ in the experimental group vs. >290 in the control) and a rise in performance consistency ($IQR = 90-99$ vs. $76-98$), which means that more students are involved in a way that lasts and is consistent. Nonetheless, this interpretation is speculative due to the absence of clear measurements for autonomy, competence, relatedness, or cognitive effort (NASA-TLX) (Hart and Staveland, 1988; Paas *et al.*, 2003), indicating a need for more study.

These results should be regarded with caution. The effect size is statistically significant but tiny ($\eta^2 = .061$), and the cross-cohort design may provide challenges. Differences across semesters (in terms of student demographics, outside factors, and scheduling issues) may affect the findings, no matter what framework is used. Most crucially, artifact quality evaluations alone do not directly assess the psychological attributes posited by the framework, including cognitive load, intrinsic motivation, or the transfer of learning. We need more tools to validate these intermediate processes.

These early results provide preliminary support to Hypotheses 2 (Intrinsic Motivation) and 4 (Transfer of Learning), as seen by higher and more consistent artifact quality ratings. Nonetheless, many significant drawbacks need recognition:

1. Single-institution design: Results pertain only to engineering education; extrapolation to other settings needs more investigation.
2. Cross-cohort confounding: Variations across semesters (schedule, student demographics, external influences) may account for the observed discrepancies.
3. Incomplete measurement: Structured Academic Activities scores alone do not directly assess cognitive load (via NASA-TLX or other instruments), intrinsic motivation (using the SDT scale), or transferability (through innovative problem-solving activities). These approved tools must be used in future studies.
4. Absence of comparison on conventional metrics: Although the framework preserves test scores, a direct comparison of exam performance is essential to substantiate our assertion that TISE-VALORIZE does not undermine traditional results.

Post-hoc power analysis (using G*Power 3.1, $\alpha = 0.05$, observed $\eta^2 = 0.061$, two-tailed test) shows that the power for the Experiment vs. Control 2 comparison is around 0.68, which is below the usual 0.80 level. According to Cohen's rules, this reported η^2 is a small to medium effect size ($\eta^2 = 0.06$). This emphasis indicates that the present research may not possess the power to consistently identify minor effects in independent samples. Subsequent investigations should aim for a sample size of $N \geq 200-250$ per group to get a statistical power of ≥ 0.80 for a modest to medium effect size ($\eta^2 \sim 0.06-0.10$).

A forthcoming empirical agenda for stringent validation necessitates: validated instruments for cognitive load (CLT-based measures), intrinsic motivation (SDT-aligned scales), and professional

identity (self-efficacy inventories, ethical reasoning); matched or randomized cohorts with pre-post designs; multi-institutional studies to assess generalizability; longitudinal tracking of artifact quality, reuse, and community value; and qualitative phenomenological studies of student and educator experiences.

6 Discussion: Theoretical Implications and Addressing Counterarguments

5 The present study does not directly test all proposed mechanisms of the TISE-VALORIZE framework. Instead, it offers a preliminary empirical indication that the framework may be associated with higher and more consistent performance outcomes, while the proposed motivational, cognitive, and identity-related mechanisms remain to be tested in future studies. Practically, the framework suggests that engineering educators can redesign courses around structured knowledge mapping, professional simulation, and AI-supported feedback to improve consistency of student outcomes.

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5.16.1 How TISE-VALORIZE Follows the Principles of CLT and SDT in Learning Science Interpretation of Findings

The present study provides preliminary empirical evidence that the TISE-VALORIZE framework may be associated with higher and more consistent performance outcomes in a digital engineering education context. The intervention group showed both higher mean scores and lower score dispersion than the two conventional instruction groups. However, the observed effect size was modest, which suggests that the framework may have contributed to performance differences under the conditions of this study, but it does not account for most of the observed variance. This framework combines important concepts from contemporary scientific education. It uses Cognitive Load Theory and a dual knowledge mapping method to help people organize information and cut down on needless work. The transition from Primitive Maps, which establish foundational knowledge, to Problem-Solving Maps, which cultivate practical abilities, aligns with the successful scaffolding shown in CLT research (Duran, Zavgorodniia and Sorva, 2022; Robins, 2022). The framework also uses Self-Determination Theory by encouraging independence, competence, and connection via the Knowledge Marketplace. This method lets you choose, shows you how to improve your skills, and shows you how to join the community. Connecting the payment system to Bloom's Taxonomy also makes it evident how to go forward, which boosts a feeling of competence.

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These findings are broadly consistent with the instructional logic of the framework. In particular, the use of structured knowledge mapping, professional simulation, and contribution-oriented learning activities may have supported more stable performance across students. At the same time, the present study does not directly test the proposed motivational, cognitive-load, or identity-related mechanisms. Therefore, these findings should be interpreted as preliminary performance-based evidence rather than direct confirmation of the full theoretical model.

5.26.2 Concerns about AI Integration: Dependence versus Partnership Comparison with Prior Work

The present findings are broadly aligned with prior research suggesting that structured scaffolding and meaningful participation can support learning in higher education. From a Cognitive Load Theory perspective, the framework is consistent with the use of structured representations to support schema development. From a Self-Determination Theory perspective, the framework is also consistent with the idea that meaningful contribution, perceived choice, and visible progress may support engagement. However, because these mechanisms were not directly measured here, the present study should be understood as conceptually aligned with prior theory rather than as a direct empirical test of those

theories One big worry is that employing AI in the classroom might make kids rely on it instead of learning how to do things on their own. If you don't employ AI wisely, this danger increases higher. TISE VALORIZE handles this by making it explicit what AI can and can't accomplish. AI performs easy tasks, rapidly retrieves information, gives initial responses, and advises what to do next. Students still do more essential things, such establishing objectives, checking to see whether outcomes fit ideals, and thinking about what they learned. This manner, students regard AI as a useful tool for tasks, but not as a method to replace their own thinking. Also, learning diaries and the professional simulation setting lets students think about how much they rely on tools, think about other options, and learn to make their own decisions. The objective is to cultivate students proficient in collaborating with AI, using its functionalities while preserving critical judgment, a meta-skill that is becoming more vital in professional activity (Ninghardjanti *et al.*, 2025).

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6.3 Some people may think that such a broad framework is too hard to use on a large scale, yet technology, particularly AI, is made to function on a large scale. In big classrooms, teachers frequently must choose between offering less feedback to keep their workload manageable, which may hamper learning, or giving everyone thorough comments, which can lead to fatigue. With AI in VALORIZE, there is a third option: AI can provide students with feedback on a regular basis, on time, and in a way that is easy for them to understand. This lets teachers concentrate on mentoring and grading activities that need a human touch. The framework also encourages students to assist each other and work together to learn, which helps distribute the task of gaining knowledge throughout the group. There are equity issues since this system needs students to have access to the internet and digital gadgets. A lot of the technologies used, such Miro, Google Workspace, and GitHub, are still free or cheap. If a higher education institution use VALORIZE, it should make sure that all its students can utilize gadgets, the internet, and get help with technology. This is the institution's fault, not a deficiency in the system.

Limitations: This study has several important limitations. First, the design was a posttest-only quasi-experimental comparison with non-equivalent cross-cohort groups rather than a randomized controlled evaluation. As a result, cohort effects and semester-level confounding cannot be ruled out. Second, no pretest or baseline covariates were available, which limits causal interpretation and prevented the use of ANCOVA. Third, the empirical analysis relied on performance outcomes from Structured Academic Activities and did not directly measure cognitive load, intrinsic motivation, professional identity, or transfer. Fourth, demographic variables suitable for controlled comparison were not systematically available. Fifth, formal inter-rater reliability and psychometric validation evidence for the rubric were not calculated retrospectively. These limitations mean that the findings should be interpreted as preliminary and performance focused.

6.4 Future Directions

Future studies should test the framework using stronger designs, including matched or randomized groups, pretest-posttest comparisons, and larger multi-institutional samples. In addition, future work should include validated instruments for cognitive load, intrinsic motivation, professional identity, and transfer, so that the proposed mechanisms of the framework can be examined directly. Secondary indicators such as artefact quality, marketplace participation, and reuse value may also provide a richer picture of how value-oriented learning operates in practice.

6.7 Conclusion

This study proposes the TISE-VALORIZE framework as a value-oriented approach to digital engineering education that integrates structured knowledge mapping, professional simulation, and contribution-based learning. In this preliminary posttest-only quasi-experimental comparison, the intervention group showed higher mean performance and lower score dispersion than two conventional instruction groups, although the observed effect size was modest. Practically, the framework suggests a possible direction for redesigning engineering courses around structured representation, authentic participation, and AI-supported feedback. Future research should test the framework with stronger designs and direct measures of cognitive load, motivation, professional identity, and transfer. Knowledge mapping, gamification, and AI-based learning are all well-known ideas, but TISE-VALORIZE is distinct in three fundamental respects. First, it employs an integrated approach that connects cognitive science (CLT), motivation theory (SDT), and research on professional growth. Most frameworks only look at one or two of these areas. Second, TISE-VALORIZE puts professional identity at the top of its list of goals. This method is different from many AI education frameworks that

concentrate on customization or content distribution. It helps learners become professionals who build their identity, values, and ethical reasoning along with their technical abilities. Third, TISE VALORIZE's style of grading is different from giving each student a grade. Instead, it emphasizes community contributions via a knowledge marketplace. Here, the work that students do not only demonstrates what they know, but it also helps their peers and future students.

AI can figure out common engineering issues quicker and more correctly than most students. Because of this, regular tests that check problem-solving abilities are no longer a good approach to judge someone's professional talents. TISE VALORIZE is a framework that was designed to help people get the most out of these improvements. It is based on well-known learning theories including Cognitive Load Theory, Self-Determination Theory, and studies on expertise. The framework also looks at what AI and digital technologies can and cannot achieve, and it focuses on helping students construct a strong professional identity and make genuine contributions. The main idea is simple: students will be more motivated, have a stronger professional identity, be better at sharing knowledge, and work together in a real sense if they study using TISE VALORIZE principles instead of traditional methods. You may judge this by looking at how effectively groups understand and use information, how confident and focused they feel, how they see themselves as professionals, and how excellent and useful the things they produce together are. In the age of AI, this method gives engineering teachers a way to go forward. Instead, then striving to beat AI at things it does well, education may concentrate on talents that only humans have, such professional judgment, ethical reasoning, creative problem-solving, and working with AI while keeping human values in mind. Institutions that decide to make this adjustment may gain a lot. Their graduates will not only be ready for their first employment, but they will also be ready to learn new things, adapt, and make a genuine impact in a world that is being changed by AI.

78 Additional Requirements

The datasets generated for this study are available on request to the corresponding author.

89 Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

910 Author Contributions

All authors approved the final manuscript and agreed to be accountable for all aspects of the work.

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890 **12**

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Orchestrating Value Co-Creation in Digital Education Using the TISE-VALORIZE Framework: From Knowing to Becoming

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10 **Keywords:** artificial intelligence, cognitive load theory, engineering education, professional
11 identity formation, self-determination theory, TISE-VALORIZE framework, value co-creation

12 **Abstract**

13 AI can automate technical activities, but it can't teach skills that only people have. This study proposes
14 the TISE-VALORIZE framework and presents a preliminary empirical examination of its association
15 with student performance outcomes in digital engineering education. A posttest-only quasi-
16 experimental study with non-equivalent cross-cohort groups involved 138 undergraduate engineering
17 students: one intervention group implementing TISE-VALORIZE ($n = 53$) and two control groups
18 receiving conventional instruction ($n = 42$; $n = 43$). Student performance was evaluated through
19 Structured Academic Activities assessed with Bloom's Taxonomy-aligned rubric. The intervention
20 group showed higher mean scores ($M = 94.98$, $SD = 7.21$) and lower score variance ($Var = 52.01$) than
21 Control 1 ($M = 88.70$, $SD = 18.33$, $Var = 335.78$) and Control 2 ($M = 86.54$, $SD = 17.05$, $Var = 290.98$).
22 A one-way ANOVA indicated a significant group difference, $F(2, 135) = 4.41$, $p = .014$, $\eta^2 = .061$, and
23 Tukey post-hoc analysis showed a significant difference between the intervention group and Control
24 2, while the difference between the intervention group and Control 1 did not reach significance. These
25 preliminary results suggest that the framework may support more consistent learning outcomes.
26 However, direct measures of cognitive load, motivation, and related psychological processes were not
27 included in the present study. These findings provide preliminary support for the framework's potential
28 to improve performance consistency, while larger-scale studies with direct measures of cognitive and
29 motivational processes are needed.

30 **1 Introduction**

31 Engineering education is shifting from a content-based approach to competency development
32 (Charytanowicz, 2023). Other empirical studies indicate that the disparity between education and
33 industry endures, even among high-achieving graduates: analyses of software engineering practices
34 consistently reveal a misalignment between the focus of higher education programs and the
35 requirements of practitioners in professional settings, including both technical and professional aspects
36 (Garousi, Giray, and Tuzun, 2019). To build a long-lasting smart campus ecosystem, many tools and

frameworks are being created to test preparedness and include digital transformation (Imbar *et al.*, 2025). Engineering education has long emphasized foundational knowledge, routine problem solving, and performance assessment. Students study the basics, figure out how to tackle frequent situations, and test their skills with tests. High grades showed that you were ready to work in the field. But this approach included an unspoken assumption: that the mental effort needed to solve engineering issues was the same as the mental labor needed to be an engineer. This assumption is no longer valid.

Learning requires learners to organize information meaningfully and integrate it into schemas stored in long-term memory. (Sweller, Van Merriënboer, and Paas, 1998). One important thing about working memory, which is a mental region where new knowledge is kept, is that it can only contain roughly 3–7 bits of information at a time. Unless actively rehearsed, such information typically decays within 15–30 seconds (Miller, 1956). Students will have trouble learning and developing their abilities if the environment has too much information in it. Cognitive Load Theory (CLT) offers a solution: instructional design should minimize extraneous cognitive load to optimize working memory for effective learning termed 'germane load' by CLT researchers. CLT serves as a cognitive framework for crafting educational experiences that cultivate robust skills (Robins, 2022). It must alleviate superfluous cognitive burden to liberate working memory for authentic learning, and it must structure teaching to facilitate the development of meaningful, practical knowledge frameworks that embody professional competence. It must also do this in a manner that respects autonomy, competence, and relatedness that keep people interested. CLT posits that the complexity of a topic correlates with the interaction of its components. Darejeh, Marcus, and Sweller (2021) examined the impact of narrative style on cognitive load during the acquisition of a difficult productivity software program. The researchers used two categories of materials, distinguished by low and high element interaction, according to each story style.

The theoretical framework delineated below aims to actualize this possibility. It is based on well-known learning theories (Cognitive Load Theory (CLT), Self-Determination Theory (SDT), and research on expertise) and considers the realities of modern technology (AI integration, digital tools, and participatory design). Its goal is to create professionals who can lead and make a difference in a world that is changing quickly. In this preliminary study, we empirically evaluate only the performance-related dimension of the framework, while the motivational, cognitive-load, transfer, and professional-identity mechanisms remain theoretical propositions requiring direct measurement in future work. The framework draws on Cognitive Load Theory, Self-Determination Theory, and professional identity formation because these perspectives address three complementary dimensions of engineering education. Cognitive Load Theory explains how structured learning design and external representations may support schema construction and reduce unnecessary processing demands. Self-Determination Theory explains how autonomy, competence, and relatedness may support sustained engagement in learning activities. Professional identity formation helps explain why professional simulation and value-oriented participation matter in engineering education beyond content mastery alone. Together, these perspectives provide the theoretical basis for the TISE-VALORIZE framework and for its expected influence on both learning processes and performance outcomes.

In the present preliminary empirical study, two hypotheses were examined directly:

- H1. Students in the TISE-VALORIZE group will achieve higher Structured Academic Activities scores than students in the conventional instruction groups.
- H2. Students in the TISE-VALORIZE group will show lower score dispersion than students in the conventional instruction groups.

In addition, the framework proposes several secondary hypotheses for future empirical testing:

- H3. The framework may reduce perceived cognitive load through structured knowledge mapping and scaffolding.
- H4. The framework may support intrinsic motivation and learning agency through autonomy-supportive and contribution-oriented learning activities.
- H5. The framework may support stronger professional identity formation and learning transfer through professional simulation and value-oriented participation.
- H6. The framework may increase the quality and reuse value of student-produced artefacts within the learning community.

Because the present study did not include direct measures of cognitive load, motivation, professional identity, or transfer, only H1 and H2 are evaluated empirically in this paper.

2 Theoretical Foundations: Merging Cognitive Science and Pedagogy

2.1 Fundamental Ideas About Education

The framework is based on several interconnected ideas that are meant to meet the needs of the AI future. First, Value-Oriented Education (VOE). In traditional education, teachers teach technical information, and students learn it. VOE alters this goal. Higher education should teach how to be good people, how to be professionals, and how to give back to society. Students learn how to develop code and why technological choices might hurt people or the environment, for example. Second, Value Co-Creation (VCC). This word originates from marketing and service writing and talks about how businesses and customers work together to produce value. In education, VCC means that students do not merely receive knowledge passively; they actively contribute to creating learning experiences that matter. They collaborate with educators and technology to develop resources such as knowledge maps, problem-solving tools, and educational aids for their classmates and future learners. In this way, students become co-creators instead than merely consumers of knowledge (Prahalad and Ramaswamy, 2004; Grönroos and Voima, 2014; Robayo-Pinzon *et al.*, 2023).

Knowledge Management (KM) is a broad way for an organization to gather, develop, protect, share, and use knowledge gained from its experiences (Mohammed, 2021; Deev and Finogeev, 2023). It works well with the focus of VOE and VCC. Collaborative Learning (CL) is a strategy in which a small group of students work together to achieve a shared objective or complete an assignment (Ouyang, Xu and Cukurova, 2023; Ramadevi *et al.*, 2023). Lastly, Personalized Learning (PL) is a method of teaching that changes based on how fast students learn and how lessons are given to help them achieve their best (Labib, Canós, and Penadés, 2017; Deev and Finogeev, 2023; Apirating, 2025).

2.2 Psychological and Cognitive Foundations

CKMS-SE combines old ways of thinking about psychology and cognition:

- Self-Determination Theory (SDT) explains that individuals are intrinsically driven when their demands for autonomy, competence, and relatedness are satisfied (Weydner-volkmann and Bär, 2024; Annamalai and Nasor, 2025).
- Cognitive Load Theory (CLT) emphasizes that successful learning is facilitated by reducing irrelevant cognitive burden (extrinsic load) and enhancing cognitive load that aids the learning process (Alsaffar, Alfayly, and Ali, 2022; Apirating, 2025).

- People that have similar aims, interests, and skill levels make up Communities of Practice (CoP). They acquire information collaboratively by active participation and contribution as a cohesive unit (Tsang and Tsui, 2017; Taft, 2019).

2.2.1 Self-Determination Theory and Intrinsic Motivation

CLT elucidates the cognitive frameworks integral to the learning process, while SDT emphasizes the motivating factors that sustain student engagement and facilitate profound learning (Ninghardjanti *et al.*, 2025). SDT says that human motivation exists on a scale. At one end is external regulation, where people do things because they feel pressure or get benefits from outside sources. At the other end is integrated regulation, where people do things that are in line with their own beliefs and identities. SDT also makes a distinction between the amount of motivation (how strongly someone feels motivated) and the kind of motivation (the qualities or categories of motivation they have).

Students who still use conventional learning approaches tend to depend on grades or test results as rewards. They usually use shallow learning tactics, stop when the benefits stop, and say they are less happy. Conversely, learners driven by intrinsic factors—such as curiosity, autonomy, and the aspiration to make a significant impact, employ more profound learning strategies, maintain persistence despite the lack of external incentives, and experience enhanced well-being and continuous engagement.

2.2.2 Cognitive Load Theory and Schema Construction

New information is first processed in working memory. Working memory has severe capacity and duration constraints. (Duran, Zavgorodniaia, and Sorva, 2022). Learning is the process of transferring information from working memory to long-term memory, where it is organized into mental schemas that integrate multiple elements into a coherent whole. The creation and automation of such schemas are central to expertise development (Robins, 2022).

This research established the VALORIZE strategy, which aims to design learning that alleviates the three forms of cognitive burden on students and enhances information processing in their working memory, drawing on cognitive load theory (CLT) and dual coding theory (DCT). Students convert the signal and system domains, which were initially expressed in words and arithmetic equations, into knowledge maps and problem-solving maps.

The following design principle is clear: to make learning as effective as possible, instructional designers should reduce extraneous load, manage intrinsic load correctly (by providing scaffolding for difficult material), and increase germane load, which is the amount of working memory used for productive learning activities (Duran, Zavgorodniaia, and Sorva, 2022).

2.2.3 Community of Practice: Professional Identity and Expertise Development

The idea of professional identity, which is taking on the responsibilities, attitudes, and behaviors that come with a job, is an important part of the framework. Professional identity isn't just something you get by learning new things. It develops over time as you do real professional work, get advice from experienced professionals, and gradually adopt professional standards and values (Pischetola, 2021; Shal *et al.*, 2026).

Moreover, ethical thinking and responsible professional judgment are fundamental to professional identity (Große-Bölting *et al.*, 2023). Engineers do not work in a vacuum; their choices have an impact on people's health, the environment, and social justice (Cortázar *et al.*, 2024). A professional identity

must include a dedication to ethical ideals (Szynkiewicz, 2025) and the ability to maneuver through intricate scenarios where technical and ethical factors converge (Johannsen, 2026).

3 The TISE Architecture: Operationalizing Multifaceted Intelligence

In this paper, TISE-VALORIZE refers to the overarching theoretical framework, while CKMS-SE refers to its system-level implementation architecture. The suggested way to address the educational gap is the Collaborative Knowledge Management System for Smart Engineering (CKMS-SE). CKMS-SE turns the TISE-VALORIZE theory into a real way to solve problems that use human, artificial, and natural intelligence in students' everyday learning activities. The approach is based on dual-engine architecture. The first engine, called the PUDAL engine, organizes students' thinking into repeated cycles of Perceive–Understand–Design–Act–Learn (PUDAL). The second engine, called the PSKVE engine, manages a multi-asset value portfolio (Product, Service, Knowledge, Values, Environment) to find and reward multi-dimensional value co-creation.

3.1 Defining the Triune-Intelligence Model

Triune-Intelligence Smart Engineering (TISE) proposes a method for intelligent problem-solving that integrates three complementary forms of intelligence: Human intelligence distinguishes individuals by their capacity for moral reasoning, alignment of values, strategic thinking, innovative solution generation, and comprehension of contextual nuances. Artificial intelligence is the capacity to swiftly see patterns, make things better, put together a lot of knowledge, and follow computer instructions that are clear. AI systems are fantastic at working with large amounts of data, seeing minor patterns, and repeating the same tasks again with remarkable precision. Natural Intelligence: The knowledge that comes from the rules of society, the way things work in nature, and the way people think they should be. This kind of intelligence involves learning about the rules and patterns of nature, how various civilizations make sense of things, and the information that is passed down via different groups and traditions. It makes sure that new technologies are good for people and the environment in the long run.

3.2 The PUDAL Thinking Cycle as A Cognitive Method

The PUDAL engine has five steps that make up a full cycle: Perception (P): Students find stakeholders, data sources, and environmental limitations that are important to the issue. Understanding (U): They use the values at play, hidden patterns, and physical laws to get a full picture of what's going on. Solution Development (D): They look at more than one option, not simply the best one. Action (A): They put the selected solution into action by making a physical item, such code, an analytical report, a user interface, or a prototype. Learning (L): They think about how what they do affects people, computers, and the environment.

VALORIZE employs Knowledge Maps structured in a multi-graph taxonomy to make this expert cognitive process clear. This is like a "grammar of knowledge" that helps experts think. Formal knowledge artifacts back up each step of PUDAL (see Table 1). The whole process isn't a straight line; instead, the PUDAL cycle is iterative and always changing, like micro-evolution that leads to constant progress.

Table 1. PUDAL Cycle: Analysis of TISE and Artefact Requirements

PUDAL Phase	Human Intelligence Analysis Task	Artificial Intelligence Analysis Task	Nature Intelligence Analysis Task	Required VALORIZE Artefacts
Perceive (P)	Find out who the most important stakeholders are and what they think about the problem in general.	Define data sources and collection strategies.	Find the most important environmental indicators and physical limits that need to be assessed.	Factual Association Map
Understand (U)	Find out what stakeholders' value, what ethical challenges they face, and what the cultural environment is.	Analyze data to identify patterns, correlations, and barriers.	Model the laws of physics and the rules of nature that control the system.	Portfolio: Directed Acyclic Graph (DAG), Decomposition Tree, Flowchart
Decision-making and Planning (D)	Set up ethical guardrails and success metrics that are based on people.	Use optimization algorithms or simulations to evaluate solution options.	Look at your possibilities in light of your resources and how well they work.	Trade-Off Analysis Map (Weighted Graph)
Act-Response (A)	Make user interfaces and service delivery models that people can utilize.	Make algorithms for controlling things, pipelines for processing data, or actions for robots.	Design physical components and artefact mechanisms.	Process Application Map (Flowchart)
Learning-Evaluating (L)	Think about how the answer will affect people's health and equality in the long run.	Think about how the answer will affect people's health and equality in the long run.	Evaluate the non-negotiable final impact of the solution on the environment.	Synthesized Design Map (Network Graph)

203 **3.3 The PSKVE engine as a value-creation method**

204 The Knowledge Marketplace was changed into a multi-asset PSKVE value portfolio (see Table 2).

205 **Table 2. Dimensions of PSKVE and TISE-VALORIZE Assets**

PSKVE Dimension	TISE-VALORIZE Asset	Dimension of Valued Worth	Example of Artefact Value Creation
Product (P)	Productivity Units (PU)	Economic Value (Productivity and Efficiency)	Writing Python programs to make computations easier or improve designs.
Service (S)	Service Vouchers (SV)	Human-Centered Design (Well-being and Empathy)	Do a thorough study of stakeholders or a map of the user experience.
Knowledge (K)	Knowledge Credits (KC)	Cognitive Depth (Intellectual Capacity)	Makes a formal Directed Acyclic Graph (DAG) that shows how different concepts are related to each other.
Value (V)	Economic Shares (ES)	Sustainable Exchange (Financial/Social Viability)	Come up with a workable business model or cost-benefit analysis for fair value distribution.
Environment (E)	Sustainability Bonds (SB)	Protective/Ethical Values (Risk and Sustainability)	Do a life cycle evaluation or create systems that really reduce their influence on the environment.

206 **3.4 Operationalization through the Knowledge Marketplace and Knowledge Maps**

207 The method uses two main tools to embed the dual engines into daily learning: (1) a knowledge
208 Marketplace, where lecturers post weekly "learning needs" as sprint goals and students voluntarily
209 respond by making PSKVE-typed artefacts; and (2) AI-assisted Knowledge Maps will help students
210 express their thoughts using the formal multi-graph taxonomy, like Directed Acyclic Graphs (DAGs),
211 flowcharts, and weighted graphs. Git and GitHub make it simple to maintain track of all their assets
212 and make sure that contributions are clear, can be used again, and are straightforward to verify.

213 4 VALORIZE Learning: A Participatory Pedagogy for Professional Formation

214 4.1 Core Principles and the Change from Knowing to Being a Professional

215 VALORIZE fundamentally shifts the focus of education from content mastery, which is learning and
216 repeating facts and procedures, to professional formation, which is developing the intellectual, ethical,
217 and social traits that make a person a good, responsible professional. This change isn't only a change
218 in words. It has big effects on how teachers and students are responsible for their own learning, how
219 they teach, and how they grade. In a content-based learning model, pupils are seen as individuals who
220 acquire knowledge. The curriculum includes things that need to be taught, and the teachers are
221 professionals who offer that knowledge. How effectively pupils can repeat that material is how success
222 is measured. In a professional development model, on the other hand, students are seen as future
223 professionals who are doing real work. The curriculum includes things that happen in the workplace,
224 and teachers act as mentors to help students grow professionally. Success is based on the ability to add
225 value and contribute.

226 4.2 Professional Simulation as Pedagogical Strategy

227 The lecturer's job varies a lot. The lecturer is no longer merely a "sage on the stage" who shares
228 information. Now, they are "representative and role models of the profession." This isn't just a shift in
229 words. There are real-world effects: the lecturer demonstrates how to handle difficulties like a pro by
230 stating their ideas out loud and making their unspoken knowledge evident. The teacher teaches students
231 how to be professionals by providing them feedback that goes beyond merely being accurate and looks
232 at aspects like how well they match with professional standards, quality, and depth.
233 Professional values are not only an additional feature of the VALORIZE paradigm; they are the most
234 significant part of any learning activity. When students create control systems, they must not only
235 improve their technical performance but also consider their impact on energy consumption, user-
236 friendliness, and the environment. Learning sends a clear message: being good at something and being
237 ethically responsible are two sides of the same coin.

238 4.3 The Knowledge Marketplace: An Economic Model for Motivation and Value Creation

239 The knowledge map is a mental structure that helps people learn, while the Knowledge Marketplace is
240 a social and motivating framework. This marketplace changes the way students, teachers, and
241 knowledge interact with each other. Instead of just following rules and taking in information, they are
242 now creating and sharing it.

243 4.3.1 How it works and why

244 The Knowledge Marketplace functions like a real marketplace. Teachers publish "learning needs" as
245 knowledge assignments (such as building a signal-system concept map), and students respond by
246 generating high-quality artifacts, like knowledge maps or problem-solving tools. Then, students may
247 use digital money to "buy" artifacts that match the requirements. These artifacts are enduring resources
248 that future organizations may use to learn from and build on. This sets off a virtuous cycle: when
249 students' effort adds value, they receive higher marks, and their classmates have access to richer
250 learning resources.

251 Teachers need to ask for artifacts first. They normally put them into groups based on how hard they
252 are to think about and the technical field. Then, students make Primitive Maps and Problem-Solving
253 Models that are good for teaching. It takes a lot of time and thinking to become an expert at these.
254 Next, the artifacts are evaluated to see whether they are comprehensive, clear, and in line with what

experts know. People who pass the examinations may "buy" them via a tiered digital currency system. The course's Knowledge Management System releases a few artifacts. These artifacts become permanent resources that present and future students may use and edit, which helps the community learn more. Lastly, pupils' final marks depend on how much money they gain by selling artifacts. This makes it clear that performing well in higher education is directly related to being able to undertake important intellectual work. It also promotes intrinsic motivation (since the activity is regarded as benefiting the community), competency via explicit feedback, and autonomy because students may pick what objects they build and how deep their labor is.

4.3.2 Gamifying Cognition: Aligning Incentives with Bloom's Taxonomy

The Knowledge Marketplace has a tiered currency system that is meant to encourage deeper study. Each level is closely related to Bloom's Taxonomy. At the basic level (Money Points, Levels 1–2), students learn the basics by making clear and correct idea maps and systems. When students reach the application level (Gold Points, Level 3), they must show how these ideas operate in actual engineering projects, such as making a filter for an audio system. They look at numerous ways to solve a problem and talk about the pros and cons of each at the analysis and assessment level (Platinum Points, Levels 4–5). At the highest level (Diamond Points, Level 6), they come up with whole new ways to solve hard and messy technical problems. This tiered framework helps students who want to get the most out of their learning naturally go from memorizing to analyzing and synthesizing.

4.4 Roles And Agency: From Consumers to Co-Designers

4.4.1 The Student as Metacognitive Agent and a Source of Value

Students in VALORIZE have a significantly different role than they do in conventional classes. They don't simply passively take in information; they actively learn and add to the group's knowledge. Creating Knowledge Actively: The major role of students is to make knowledge artifacts, like Primitive and Problem-Solving Maps, that indicate they understand the material and are useful to the course community. This is not simply a section of the test; it's the most important aspect. Assessment occurs not via high-stakes tests, but through the continual evaluation of job quality. Thinking about your own thinking: Students must maintain reflective learning journals that illustrate how they are learning. Using a framework like DAR (Description What Happened; Analysis), students learn to look at their own learning, uncover gaps, and make plans on how to improve. What caused that to happen? What does it mean to reflect? Action Plan What's next? This practice improves metacognitive awareness, which is the ability to think about how your brain works. This is important for being able to adjust. Professional Practice: The framework purposefully incorporates industry-standard protocols into student projects. For example, students may utilize version control systems like Git or GitHub to keep track of their learning artifacts, treating the process of learning as a cycle of development. This serves two purposes: it makes a detailed record of the learning process and teaches students to the professional processes they will use in real life. Portfolio Development: Instead of only being graded on their test scores, students build a portfolio of their greatest work, which they typically share on a personal blog or professional website. This portfolio becomes a real asset that shows prospective employers what you can do and immediately helps you advance in your career.

4.4.2 The Educator as Architect, Mentor, and Community Builder

As Learning Architects, teachers plan the whole learning experience, from planning weekly activities and setting requirements for the Knowledge Marketplace to sequencing tasks to build skills over time and creating social and pedagogical conditions that help students develop their professional identity. This is complicated design job that needs a lot of thought about how people learn. AI takes care of

regular tests (such as giving initial feedback and automated scoring), so teachers have more time and energy to provide good advice. This mentoring is not theoretical; teachers collaborate with students to show them how specialists solve difficult challenges. They ask deep questions to get students to think more deeply, give them feedback that helps them progress, and help each student on their way to become a skilled and moral professional. This design uses two specialized engines that work together to provide a continuous cognitive cycle inside the intelligent engineering paradigm. Figure 1. CKMS-SE System Architecture, which shows the philosophical basis of TISE-VALORIZE, the engines PUDAL and PSKVE that run the system, and the outcomes that were measured. PUDAL engine that puts into action a cycle of constant cognitive progress. This machine is a "micro-evolutionary engine" that pushes constant improvement and change. It is also a PSKVE engine that controls and improves Shared Value Creation (SVC) by making sure that the value provided extends beyond individual learning to the creation of community assets that can be verified. The Core Engine, the AI-based PUDAL Engine for Personalized Learning, and the PSKVE Engine for managing Value Co-Creation (VCC) are the three key parts of this system.

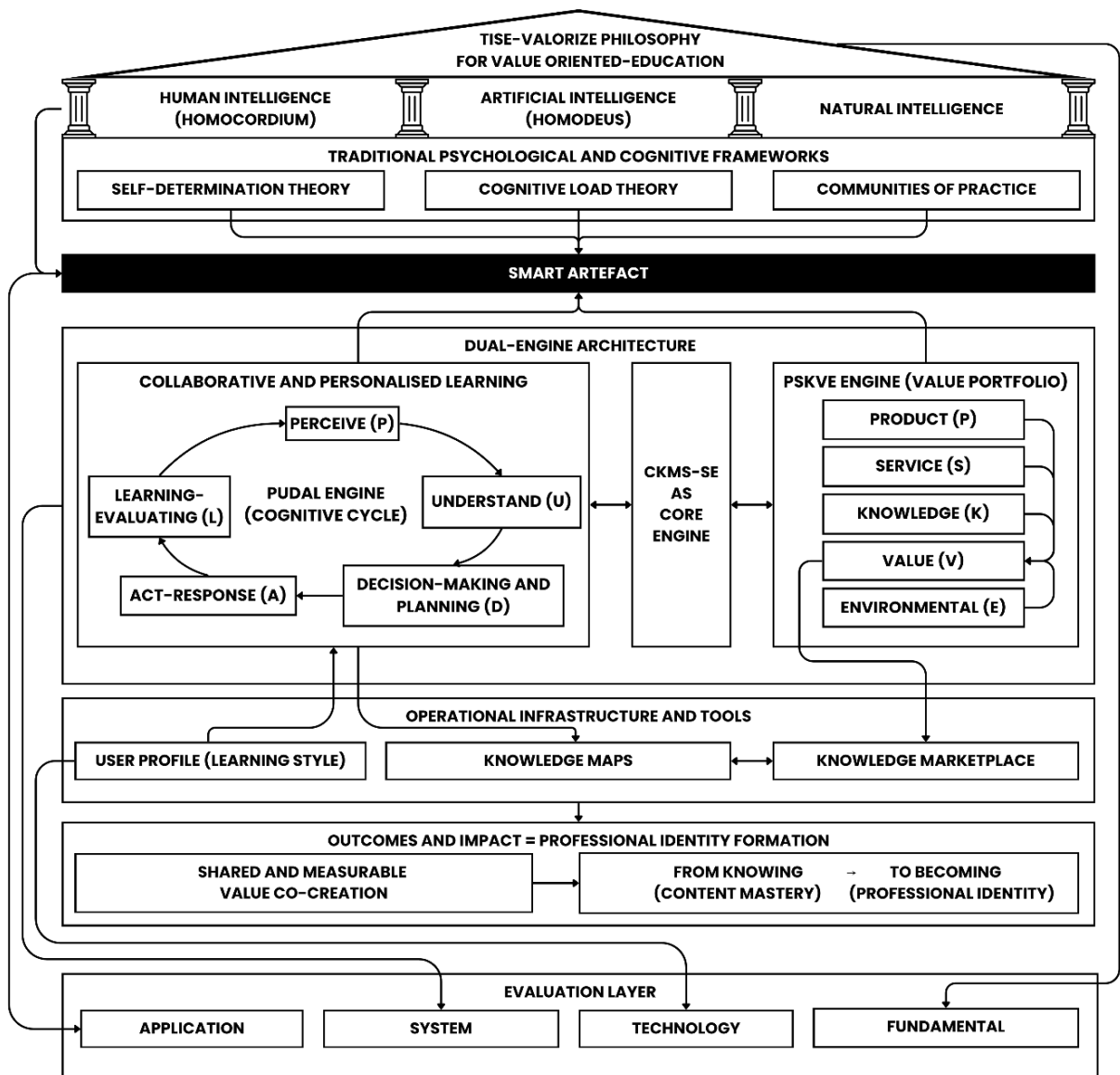


Figure 1. CKMS-SE Architecture

To clarify the empirical scope of the present study, Figure 2 distinguishes between the instructional inputs proposed by the framework, the theoretical mechanisms it draws upon, the performance outcome directly measured in this study, and the broader outcomes proposed for future empirical validation.

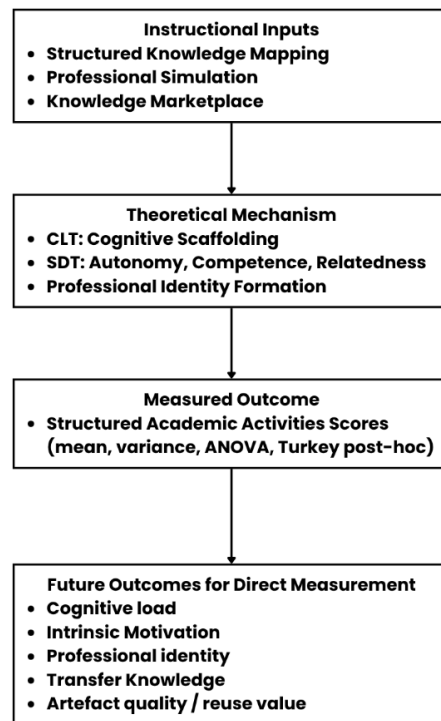


Figure 2. Conceptual scope of the TISE-VALORIZE framework in the present study

5 Preliminary Empirical Validation

5.1 Study Design

The empirical component of this paper employed a posttest-only quasi-experimental design with non-equivalent cross-cohort groups. Random assignment was not feasible because the intervention was implemented in naturally occurring course cohorts across different semesters. Accordingly, the findings should be interpreted as preliminary comparative evidence rather than causal proof.

5.2 Participants and Context

The study involved three cohorts of undergraduate engineering students enrolled in the same course context across different semesters. Control Group 1 (C1) consisted of 42 students from the even semester of 2023/2024 and received conventional instruction. Control Group 2 (C2) consisted of 43 students from the odd semester of 2024/2025 and received conventional instruction. The intervention group (E) consisted of 53 students from the odd semester of 2025/2026 and received instruction using the TISE-VALORIZE framework.

Demographic variables such as age, prior achievement, and access to technology were not systematically collected in a form suitable for retrospective controlled analysis. This limits the extent to which cohort differences can be ruled out statistically and is acknowledged as a limitation of the present study.

5.3 Assessment and Rubric

Student performance was evaluated through Structured Academic Activities using the same Bloom's Taxonomy-aligned rubric across all cohorts. The rubric emphasized conceptual clarity, mapping linkages, procedural reasoning, and integrated synthesis consistent with the expected quality of engineering problem solving. The assessed artefacts included Primitive Knowledge Maps, Problem-Solving Maps, and integrated synthesis artefacts.

Formal psychometric validation and inter-rater reliability statistics for the rubric were not calculated retrospectively in the present study. Therefore, the rubric should be understood as a common instructional assessment instrument rather than a fully validated research-scale measure. This limitation is acknowledged explicitly in the manuscript.

5.4 Statistical Analysis

The analysis included descriptive statistics, one-way ANOVA, Tukey's Honestly Significant Difference (HSD) post-hoc comparisons, Levene's test for homogeneity of variances, and Cohen's *d* for pairwise effect size interpretation. Because no pretest or baseline covariates were collected, ANCOVA could not be applied. Likewise, direct measures of cognitive load, intrinsic motivation, professional identity, and transfer were not available in the present dataset.

Before conducting the omnibus comparison, score distributions in each group were examined using the Shapiro–Wilk test. Because the score distributions deviated from normality, a Kruskal–Wallis test was also conducted as a nonparametric robustness check. One-way ANOVA was retained as the primary omnibus test for comparison with prior reporting, followed by Tukey's HSD for pairwise comparisons and Levene's test for variance differences.

5.5 Results

The intervention group showed higher mean performance ($M = 94.98$, $SD = 7.21$) than Control 1 ($M = 88.70$, $SD = 18.33$) and Control 2 ($M = 86.54$, $SD = 17.05$). The intervention group also showed substantially lower variance ($Var = 52.01$) than both control groups (C1: $Var = 335.78$; C2: $Var = 290.98$). These descriptive statistics indicate higher mean scores and lower score dispersion in the intervention group relative to both control groups. This decrease in score dispersion suggests more consistent performance in the intervention group, with fewer students located at the extremes of the score distribution. Table 3 presents the descriptive statistics for structured academic activities performance across the three groups.

Table 3. Descriptive Statistics for Scores on Structured Academic Activities

Group	N	M	SD	Var	Median	IQR	CV (%)	Min	Max
Control 1 (C1)	42	88.70	18.33	335.78	96.66	86.33-99.25	20.7	0.00	100.00
Control 2 (C2)	43	86.54	17.05	290.98	92.91	82.21-96.25	19.7	1.25	98.25
Experiment (E)	53	94.98	7.21	52.01	97.91	93.16-100.00	7.6	65.83	100.00

Note: Structured Academic Activities scores (0-100 scale); M = mean; SD = standard deviation; Var = variance; IQR = interquartile range; CV = coefficient of variation ($SD/M \times 100$); Min/Max = minimum and maximum observed scores.

The one-way ANOVA indicated a statistically significant difference in Structured Academic Activities scores across the three groups, $F(2, 135) = 4.41, p = .014, \eta^2 = .061$. Although statistically significant, the effect size was modest, indicating that group membership was associated with a limited portion of the variance in outcome scores under the conditions of this study. Shapiro–Wilk tests indicated that score distributions in all three groups deviated from normality (Control 1: $W = 0.636, p < .001$; Control 2: $W = 0.641, p < .001$; Experiment: $W = 0.736, p < .001$). Therefore, a Kruskal–Wallis test was conducted as a nonparametric robustness check. The result also indicated a significant group difference, $H(2) = 18.00, p < .001$, supporting the conclusion that score distributions differed across groups.

An important result, beyond the higher mean scores, is the substantial reduction in score variability. The intervention group showed a variance of 52.01, compared with 335.78 in Control 1 and 290.98 in Control 2. This pattern indicates lower score dispersion and more consistent performance in the intervention group, whereas the control groups showed substantially greater heterogeneity in outcomes.

Table 4. Summary of ANOVA for Structured Academic Activities scores

Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
Control (C1)	42	3725.4	88.7	335.7849463
Control (C2)	43	3721.27	86.54116279	290.9826153
Experiment (E)	53	5033.87	94.97867925	52.0050963

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	1872.578316	2	936.2891579	4.405265401	0.01401709	3.063203853
Within Groups	28692.71765	135	212.5386493			
Total	30565.29597	137				

Note: SS = sum of squares; df = degrees of freedom; MS = mean square; F = F-statistic; p = probability value; F_crit = critical F-value.

Post-hoc pairwise comparisons using Tukey’s Honestly Significant Difference (HSD) test were then conducted to examine which group differences were statistically significant, as shown in Table 5.

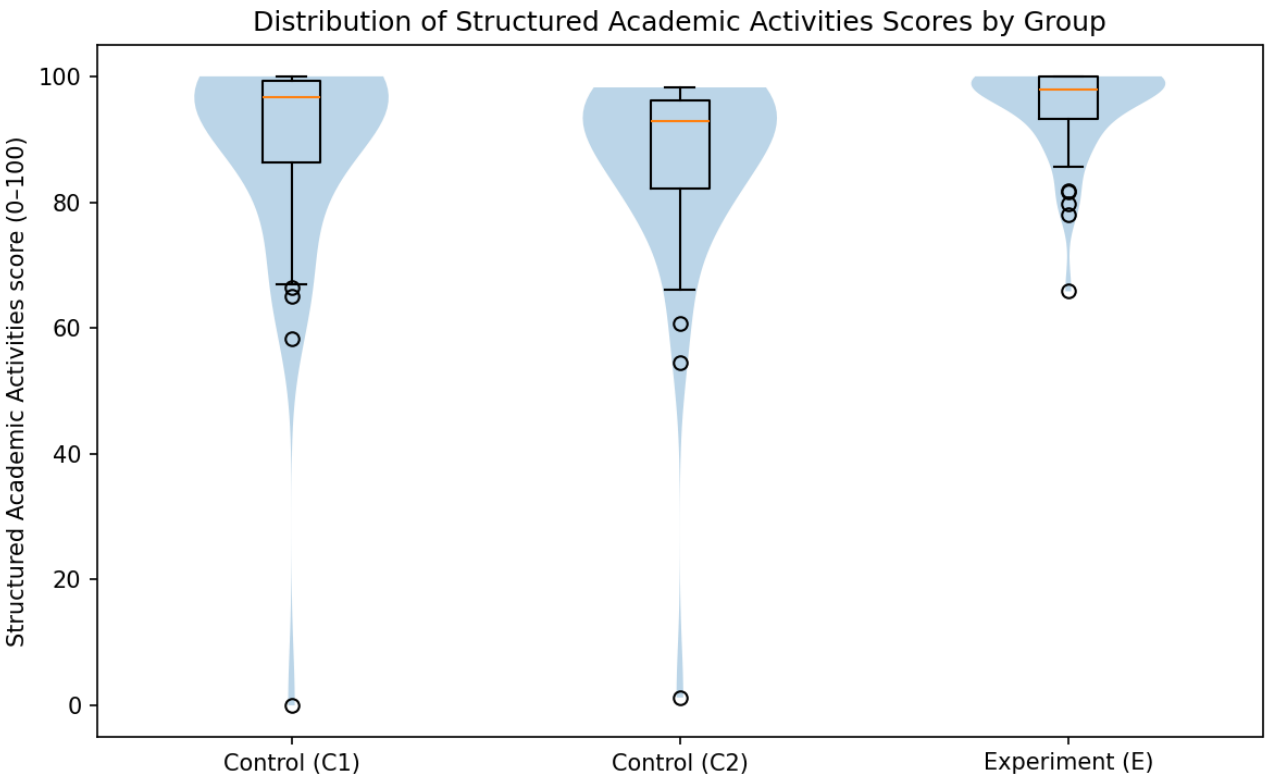
402

Table 5. Comparisons by Pair and Effect Sizes

Comparison	Mean Diff	95% CI	p-value	d (Cohen's)	Interpretation
E versus C1	6.28	[-0.86, 13.42]	.097	0.47	Not significant
E versus C2	8.44	[1.35, 15.53]	.015	0.67	Significant (medium)
C1 versus C2	-2.16	[-9.65, 5.34]	.774	0.12	Not significant

403 Note: Mean Diff= difference in group means; 95% CI = 95% confidence interval; p = probability value
404 (two-tailed); Cohen's d = standardized effect size. Significant at the $\alpha = .05$ level.

405 Figure 3 presents the score distributions across the three groups using a violin plot with box-plot
406 overlay. The intervention group shows a more concentrated distribution in the upper score range and a
407 narrower interquartile spread, whereas the control groups display broader and more dispersed score
408 distributions. This pattern is consistent with the descriptive statistics and variance analysis, which
409 suggest more stable performance in the intervention group under the conditions of this study.



410

411 Figure 3. Violin plot with box-plot overlay of Structured Academic Activities scores by group

412 Post-hoc analysis showed a significant pairwise difference between the Experiment and Control 2
413 groups (Mean Diff = 8.44, 95% CI = [1.35, 15.53], $p = .015$, Cohen's $d = 0.67$), indicating a medium
414 effect size. By contrast, the difference between the Experiment and Control 1 groups did not reach
415 statistical significance (Mean Diff = 6.28, 95% CI = [-0.86, 13.42], $p = .097$, Cohen's $d = 0.47$). There
416 was also no significant difference between Control 1 and Control 2 (Mean Diff = -2.16, 95% CI = [-

9.65, 5.34], $p = .774$, Cohen's $d = 0.12$). Levene's test for homogeneity of variances indicated that score variances differed significantly across groups, $F(2, 135) = 5.24$, $p = .006$, supporting the observation that the intervention group showed lower variability than the control groups.

This pattern of results may suggest that the adoption of TISE-VALORIZE, particularly the dual knowledge mapping system (Primitive and Problem-Solving Maps) and the Knowledge Marketplace with tiered payment incentives, produces more uniform and equitable learning outcomes. From the perspective of Cognitive Load Theory (CLT), this pattern may be broadly consistent with the intended scaffolding function of the framework. However, because cognitive load was not directly measured in the present study, no direct claim can be made regarding the framework's effect on cognitive load.

6 Discussion: Theoretical Implications and Addressing Counterarguments

The present study does not directly test all proposed mechanisms of the TISE-VALORIZE framework. Instead, it offers a preliminary empirical indication that the framework may be associated with higher and more consistent performance outcomes, while the proposed motivational, cognitive, and identity-related mechanisms remain to be tested in future studies. Practically, the framework suggests that engineering educators can redesign courses around structured knowledge mapping, professional simulation, and AI-supported feedback to improve consistency of student outcomes.

6.1 Interpretation of Findings

The present study provides preliminary empirical evidence that the TISE-VALORIZE framework may be associated with higher and more consistent performance outcomes in a digital engineering education context. The intervention group showed both higher mean scores and lower score dispersion than the two conventional instruction groups. However, the observed effect size was modest, which suggests that the framework may have contributed to performance differences under the conditions of this study, but it does not account for most of the observed variance.

These findings are broadly consistent with the instructional logic of the framework. In particular, the use of structured knowledge mapping, professional simulation, and contribution-oriented learning activities may have supported more stable performance across students. At the same time, the present study does not directly test the proposed motivational, cognitive-load, or identity-related mechanisms. Therefore, these findings should be interpreted as preliminary performance-based evidence rather than direct confirmation of the full theoretical model.

6.2 Comparison with Prior Work

The present findings are broadly aligned with prior research suggesting that structured scaffolding and meaningful participation can support learning in higher education. From a Cognitive Load Theory perspective, the framework is consistent with the use of structured representations to support schema development. From a Self-Determination Theory perspective, the framework is also consistent with the idea that meaningful contribution, perceived choice, and visible progress may support engagement. However, because these mechanisms were not directly measured here, the present study should be understood as conceptually aligned with prior theory rather than as a direct empirical test of those theories.

455 **6.3 Limitations**

456 This study has several important limitations. First, the design was a posttest-only quasi-experimental
457 comparison with non-equivalent cross-cohort groups rather than a randomized controlled evaluation.
458 As a result, cohort effects and semester-level confounding cannot be ruled out. Second, no pretest or
459 baseline covariates were available, which limits causal interpretation and prevented the use of
460 ANCOVA. Third, the empirical analysis relied on performance outcomes from Structured Academic
461 Activities and did not directly measure cognitive load, intrinsic motivation, professional identity, or
462 transfer. Fourth, demographic variables suitable for controlled comparison were not systematically
463 available. Fifth, formal inter-rater reliability and psychometric validation evidence for the rubric were
464 not calculated retrospectively. These limitations mean that the findings should be interpreted as
465 preliminary and performance focused.

466 **6.4 Future Directions**

467 Future studies should test the framework using stronger designs, including matched or randomized
468 groups, pretest-posttest comparisons, and larger multi-institutional samples. In addition, future work
469 should include validated instruments for cognitive load, intrinsic motivation, professional identity, and
470 transfer, so that the proposed mechanisms of the framework can be examined directly. Secondary
471 indicators such as artefact quality, marketplace participation, and reuse value may also provide a richer
472 picture of how value-oriented learning operates in practice.

473 **7 Conclusion**

474 This study proposes the TISE-VALORIZE framework as a value-oriented approach to digital
475 engineering education that integrates structured knowledge mapping, professional simulation, and
476 contribution-based learning. In this preliminary posttest-only quasi-experimental comparison, the
477 intervention group showed higher mean performance and lower score dispersion than two conventional
478 instruction groups, although the observed effect size was modest. Practically, the framework suggests
479 a possible direction for redesigning engineering courses around structured representation, authentic
480 participation, and AI-supported feedback. Future research should test the framework with stronger
481 designs and direct measures of cognitive load, motivation, professional identity, and transfer.

482 **8 Additional Requirements**

483 The datasets generated for this study are available on request to the corresponding author.

484 **9 Conflict of Interest**

485 The authors declare that the research was conducted in the absence of any commercial or financial
486 relationships that could be construed as a potential conflict of interest.

487 **10 Author Contributions**

488 All authors approved the final manuscript and agreed to be accountable for all aspects of the work.

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